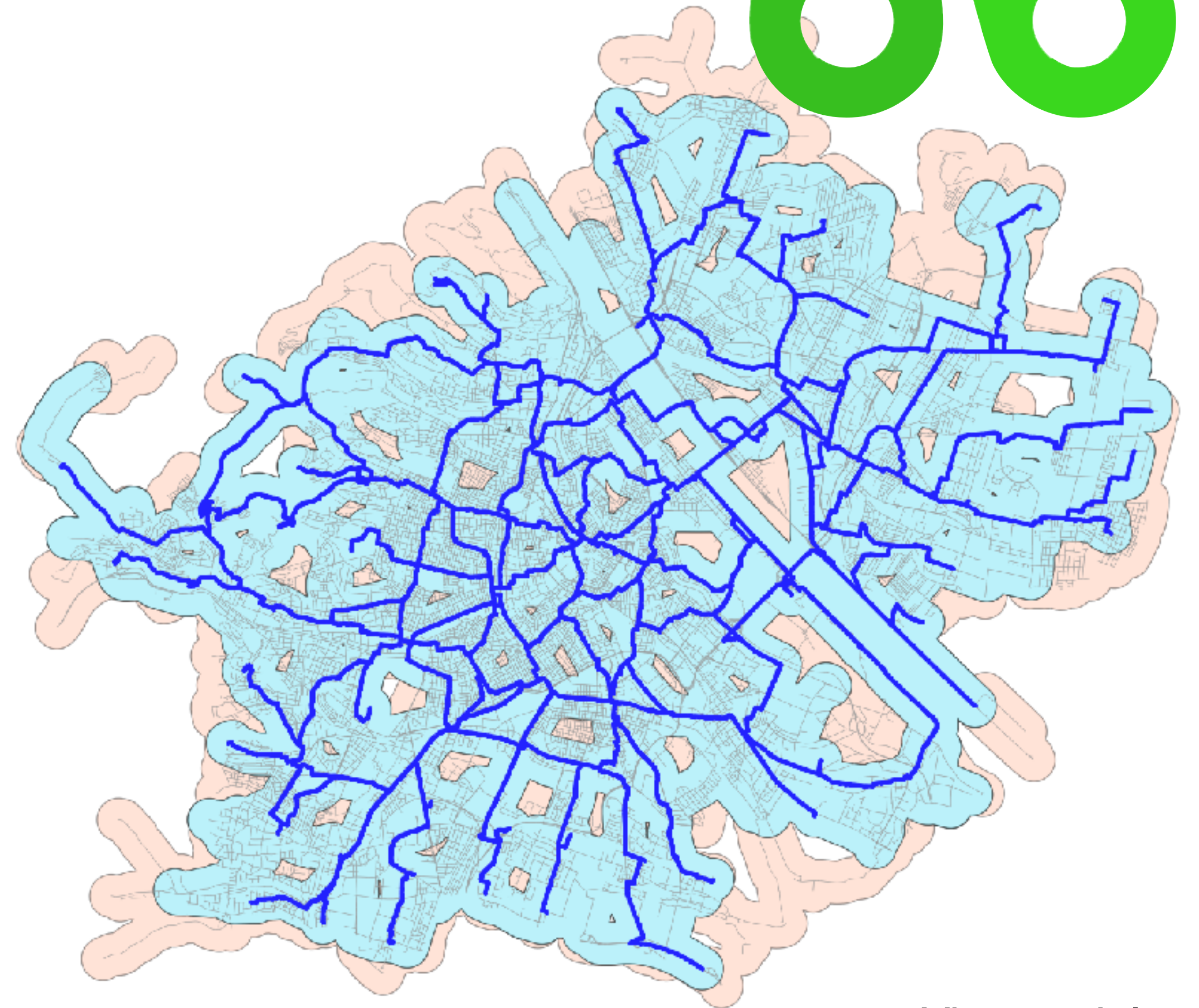
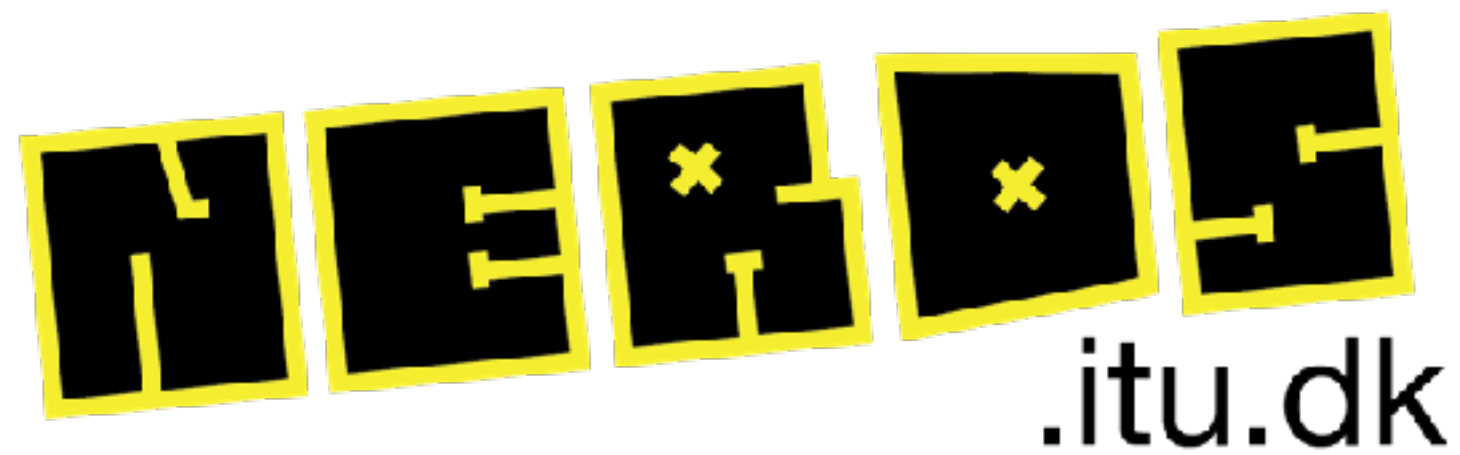


The complex socio-technical system of bicycle networks: From politics to planning software

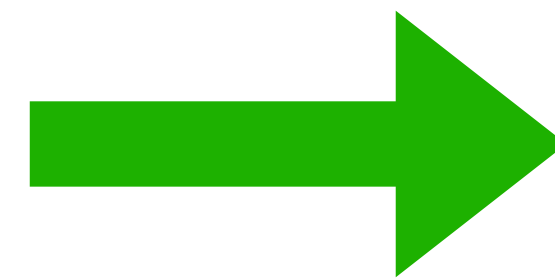
Michael Szell



IT UNIVERSITY OF COPENHAGEN

THE POTENTIAL Cycling is cost-efficient and time-tested

It solves many issues

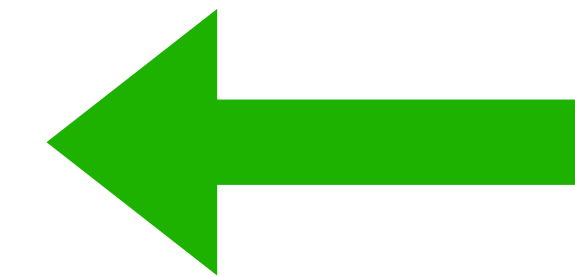


Each km travelled by Bike

brings
society
0.18 €

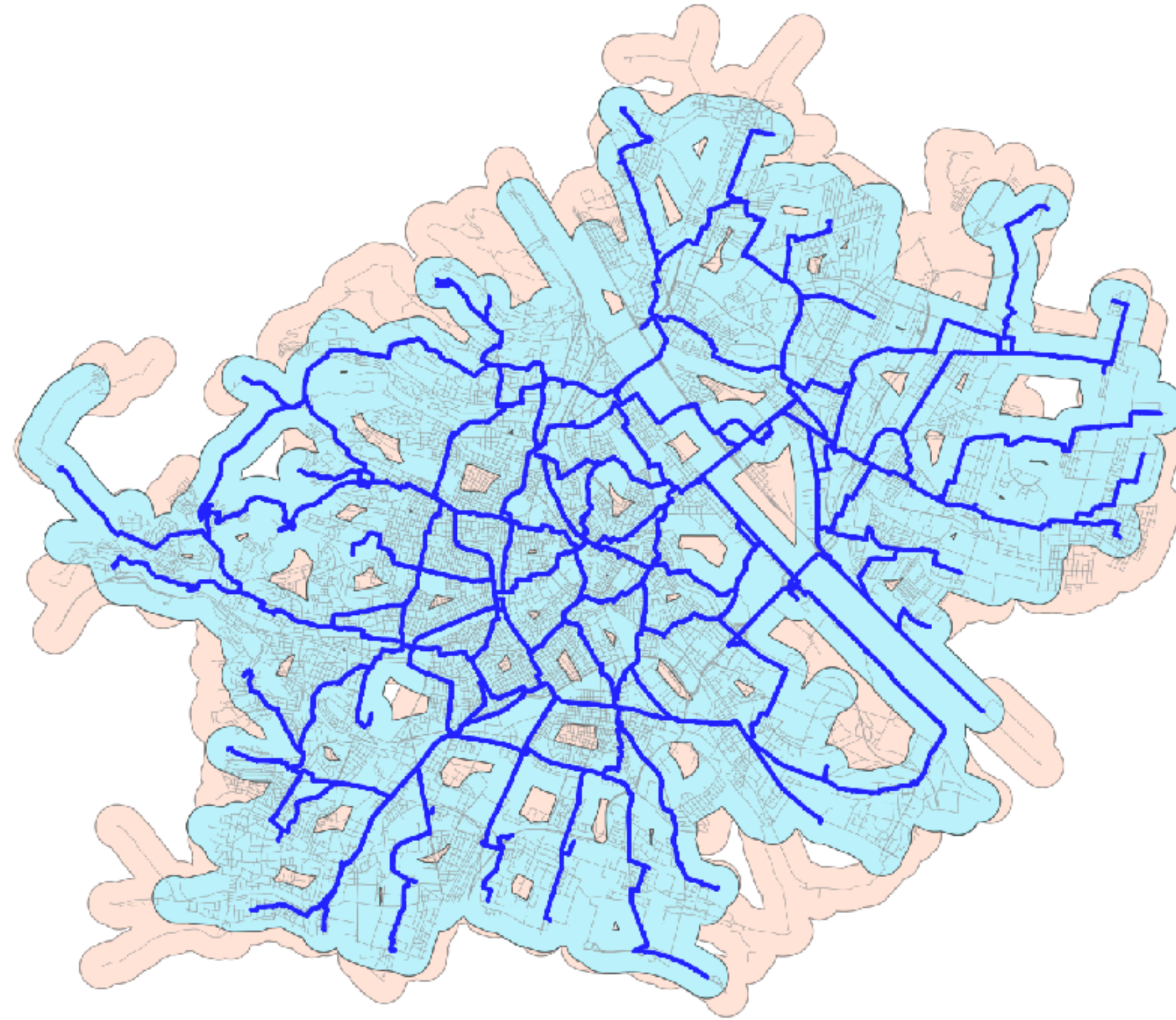
Cities gain massive
public health benefits
+ hit climate targets!

+24 Billion EUR
per year in the EU
(currently)



Gössling et al, Ecol Econ 158 (2019)
Fosgerau et al, PNAS 120.16 (2023)

THE POTENTIAL Cycling is cost-efficient and time-tested

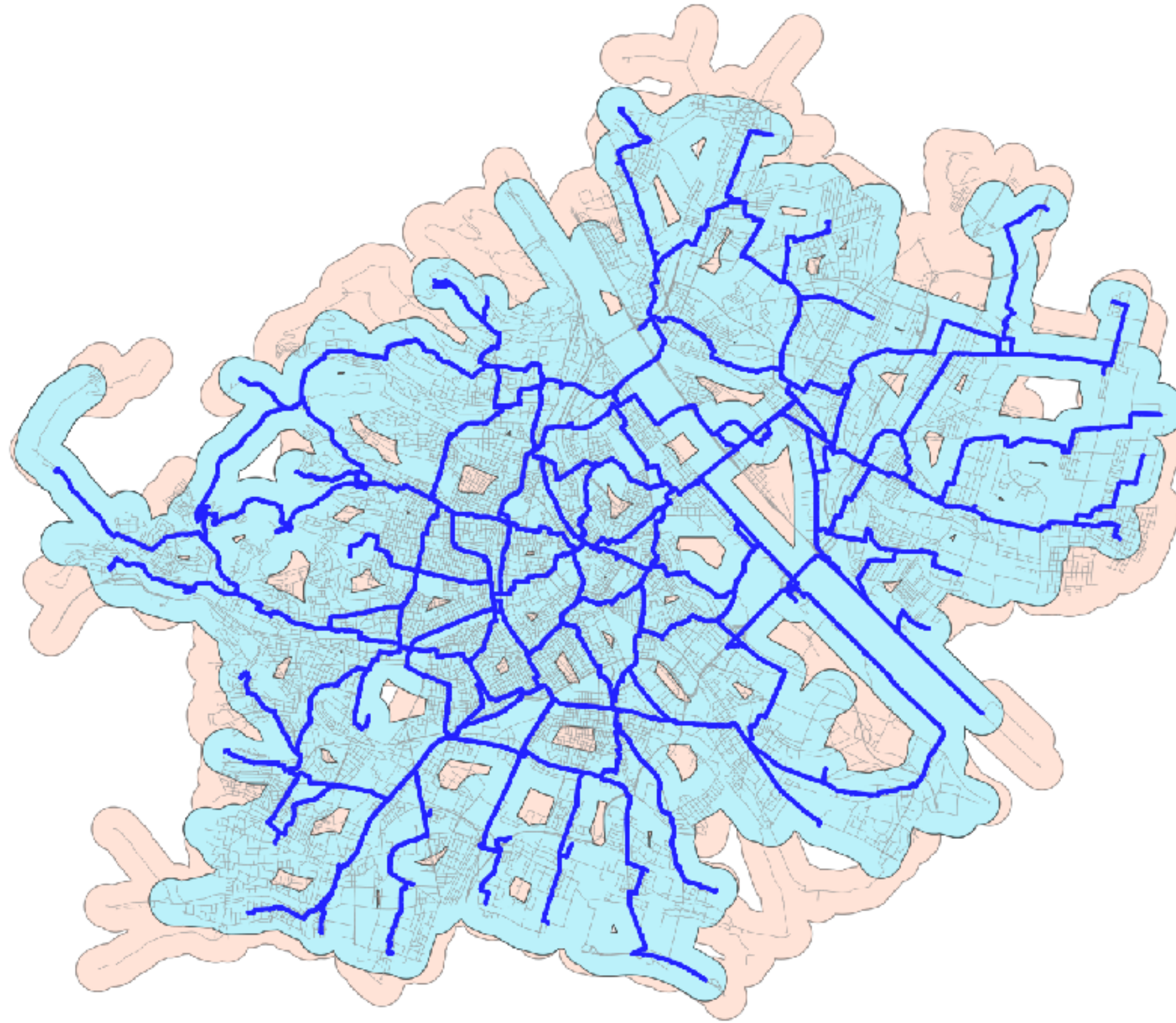


573km

>270 Mio.€

Fosgerau et al, PNAS 120.16 (2023)
Szell et al, Sci Rep 12 (2022)

THE POTENTIAL Cycling is cost-efficient and time-tested

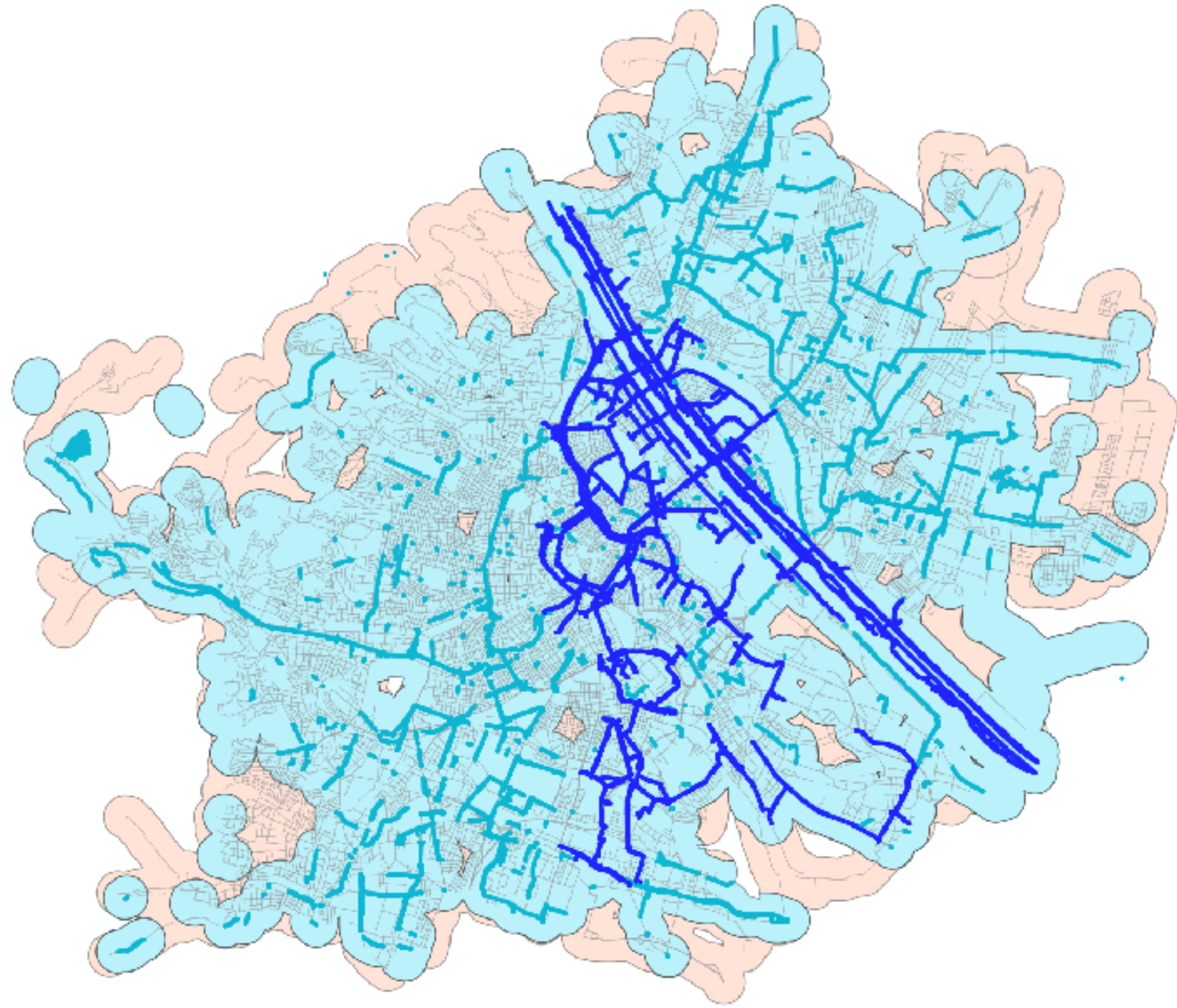


573km
>270 Mio.€

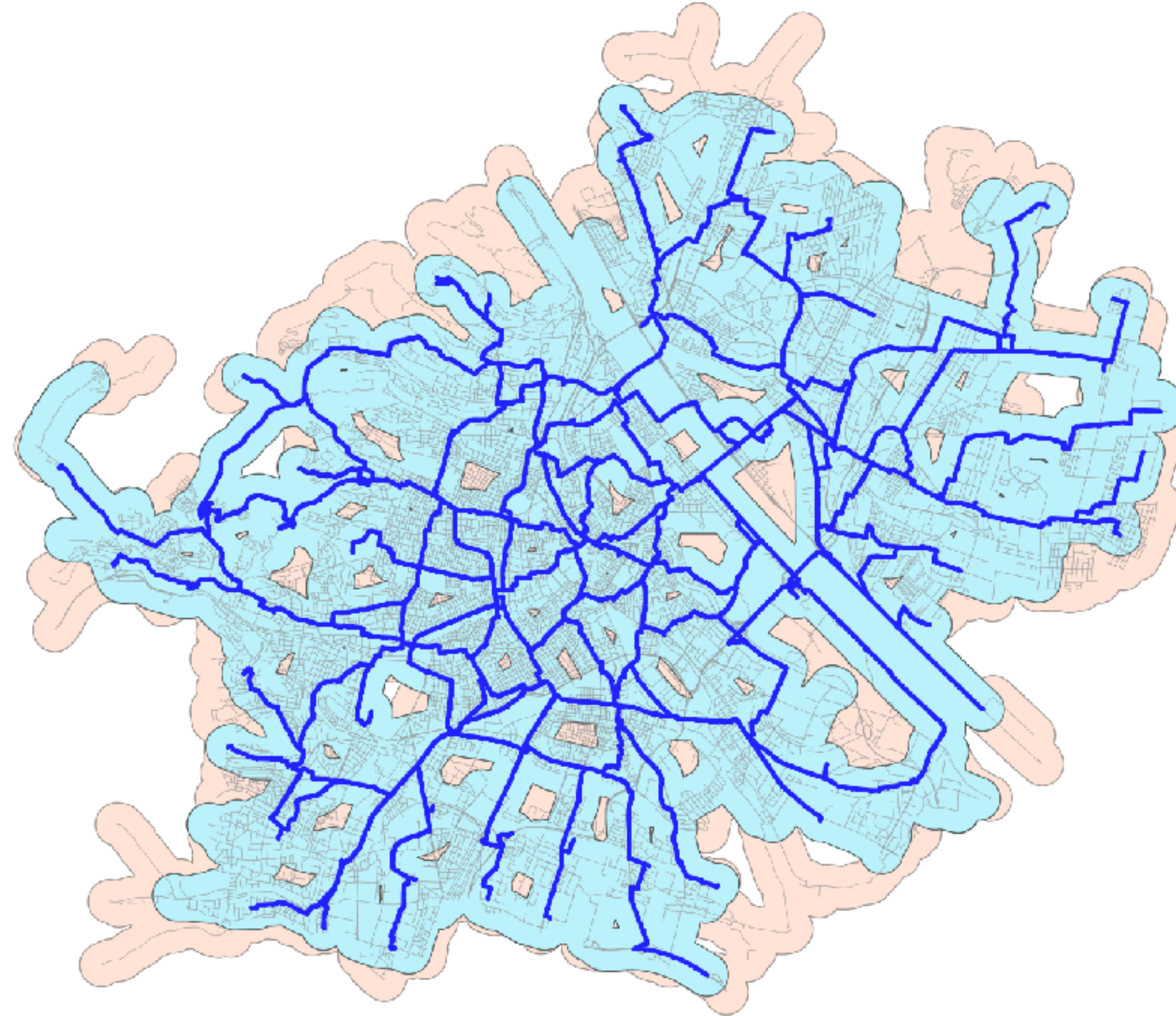
1 km of bike lane:
Breaks even after 3y
Then, +430000€ / y!

Fosgerau et al, PNAS 120.16 (2023)
Szell et al, Sci Rep 12 (2022)

THE PROBLEM Cities struggle with bike network planning



Real Vienna

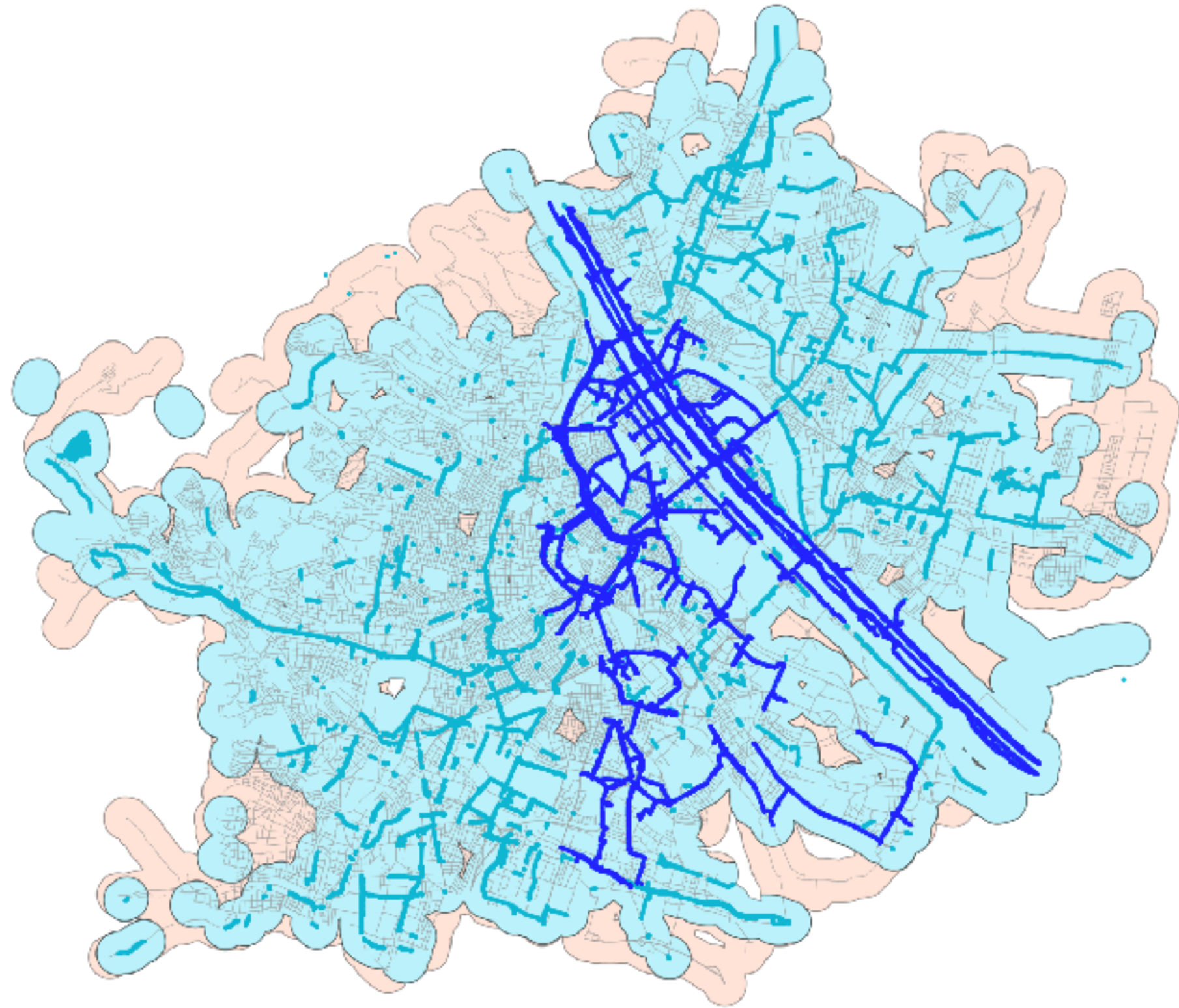


Synthetic Vienna

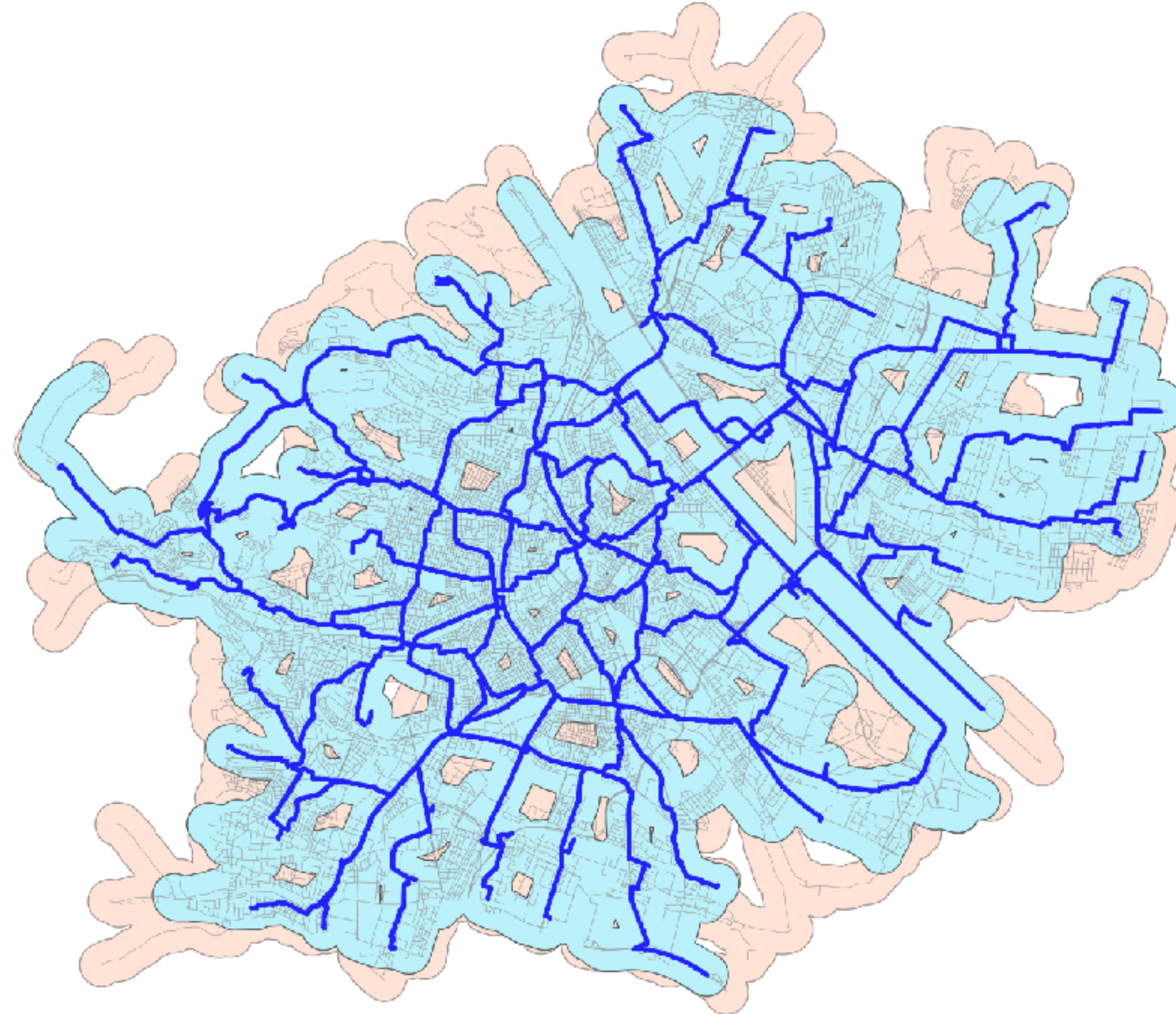
573km

>270 Mio.€

THE PROBLEM Cities struggle with bike network planning



Real Vienna

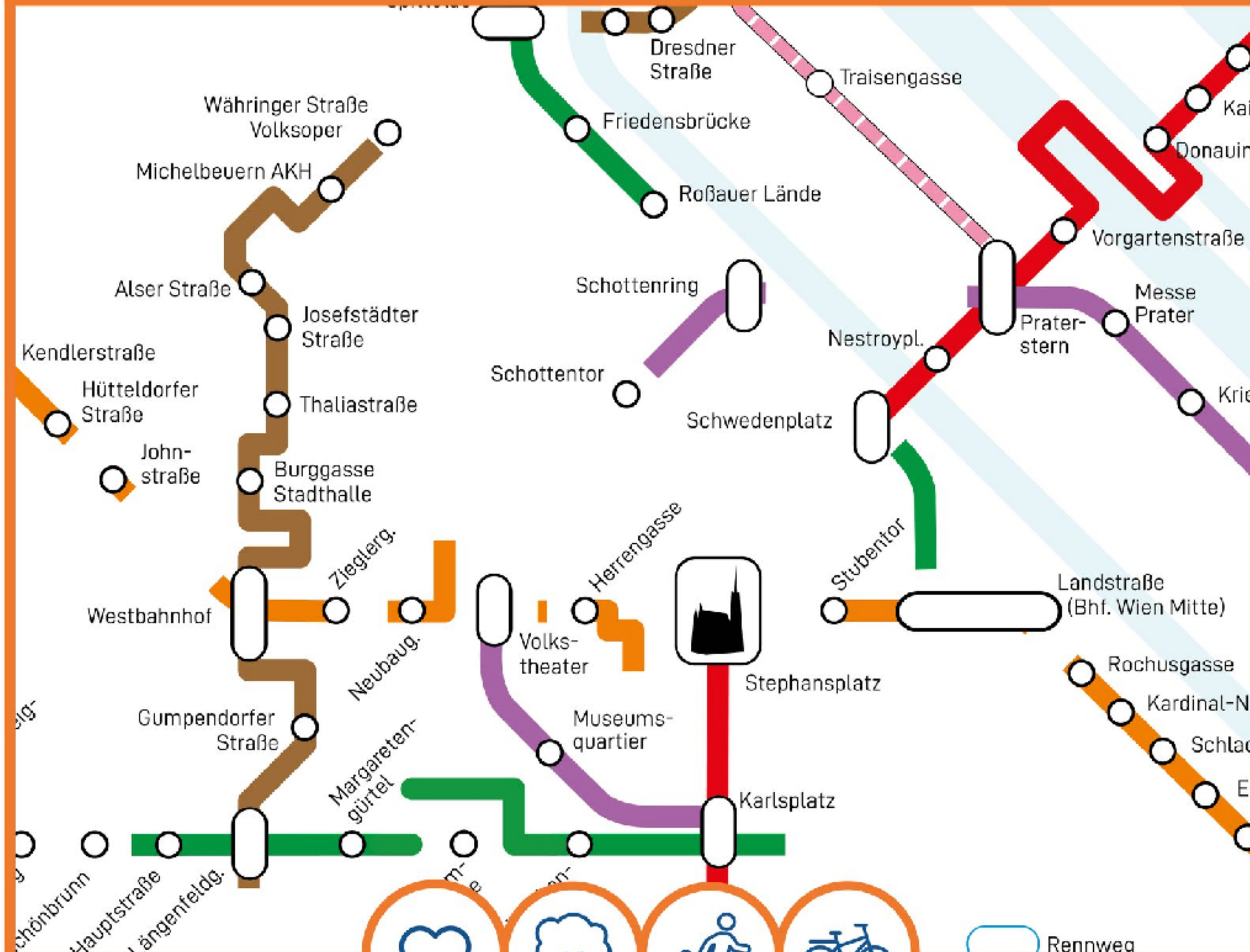


Synthetic Vienna

573km
>270 Mio.€

Our research shows:
Data-driven planning is
highly cost-efficient!

Würden wir unsere U-Bahnen bauen wie Radwege?



Quelle: Wiener Linien / eigene Bearbeitung



#PlatzFürWien

THE PROBLEM Cities struggle with bike network planning

Current network planning is:

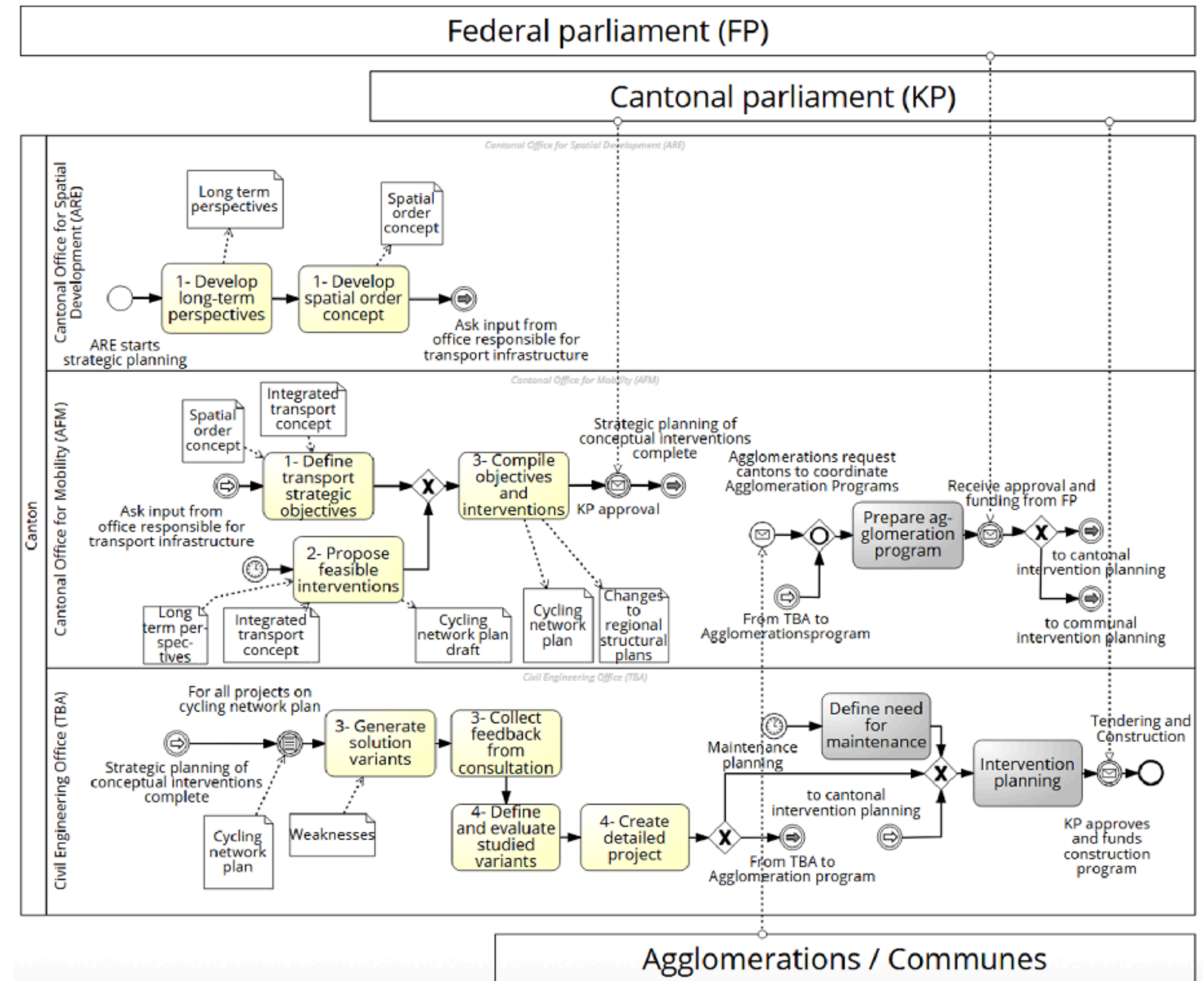
- manual
- political

THE PROBLEM Cities struggle with bike network planning

Current network planning is:

- manual
- political

Example Zürich:

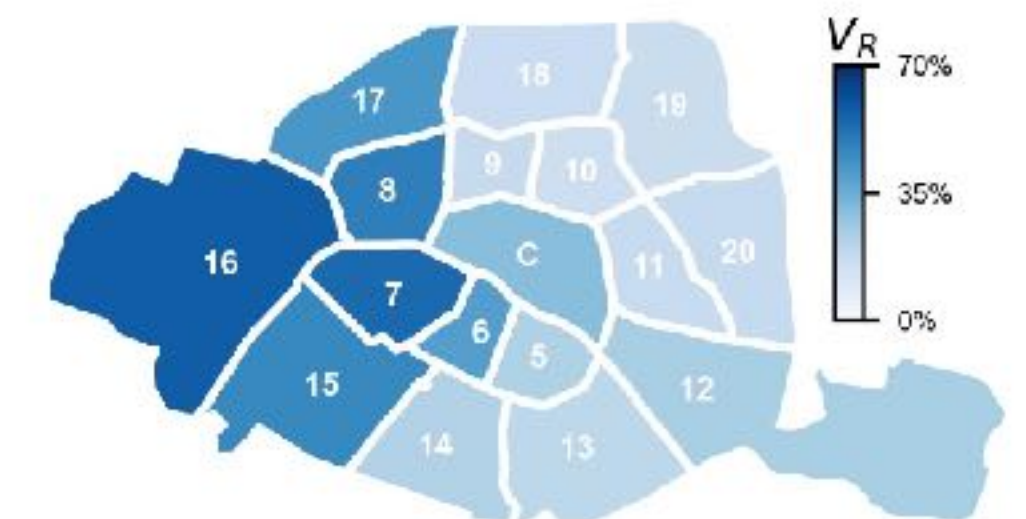
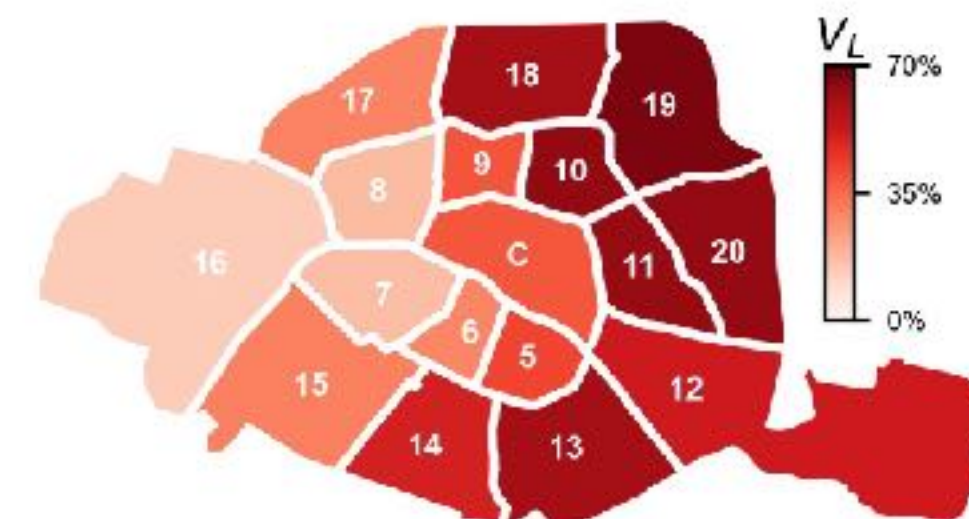
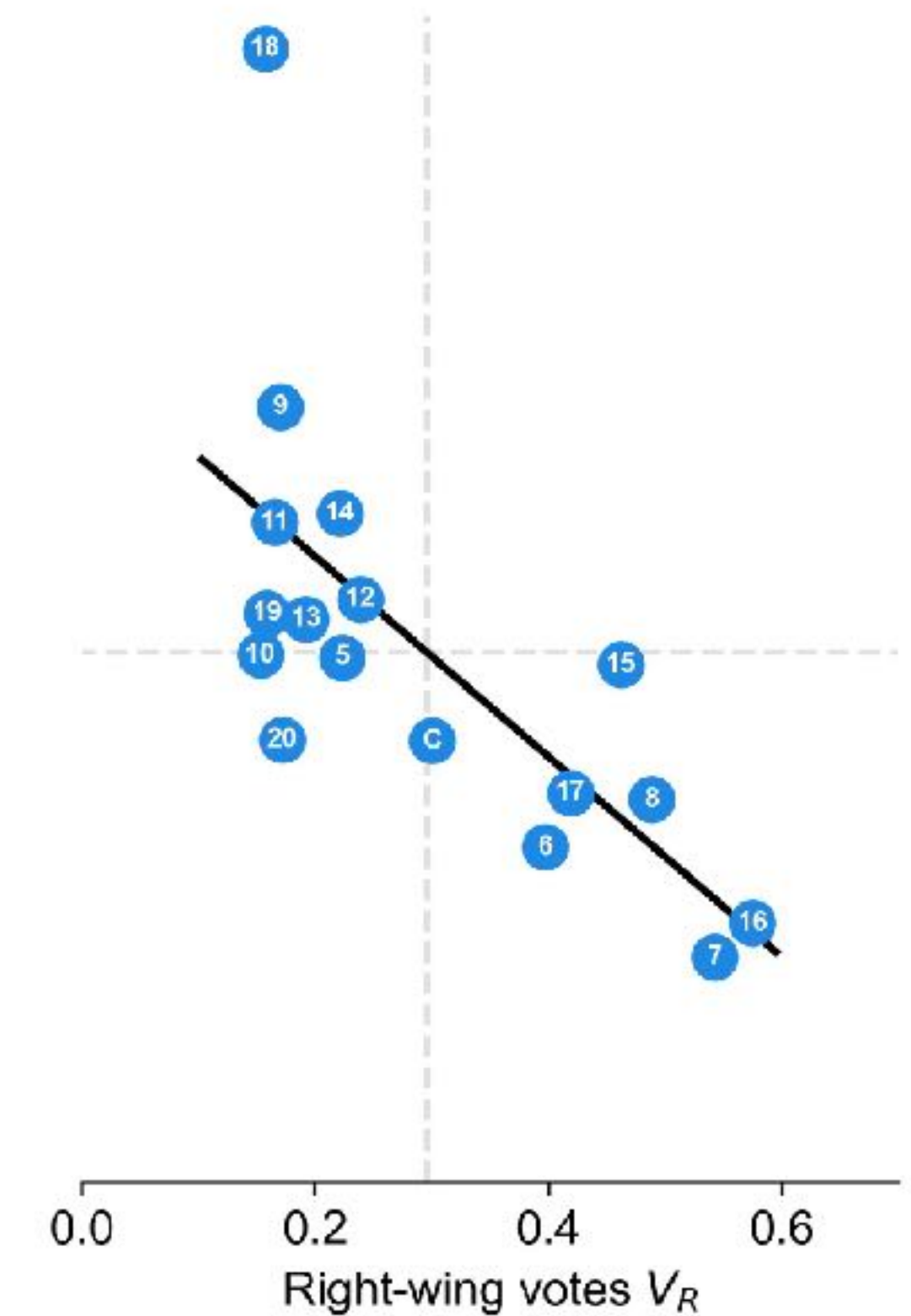
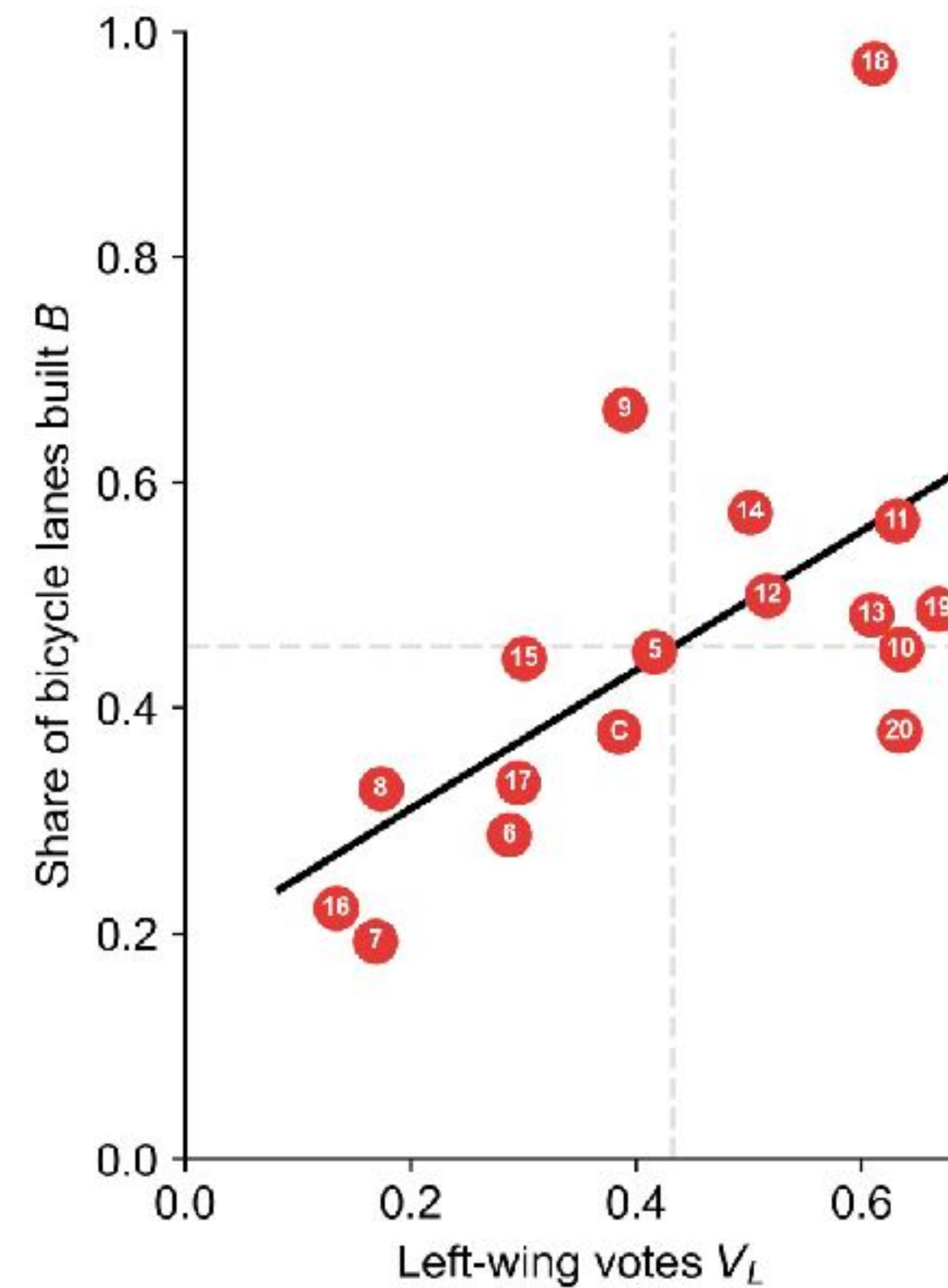


THE PROBLEM Cities struggle with bike network planning

Current network planning is:

- manual
- political

Example Paris:

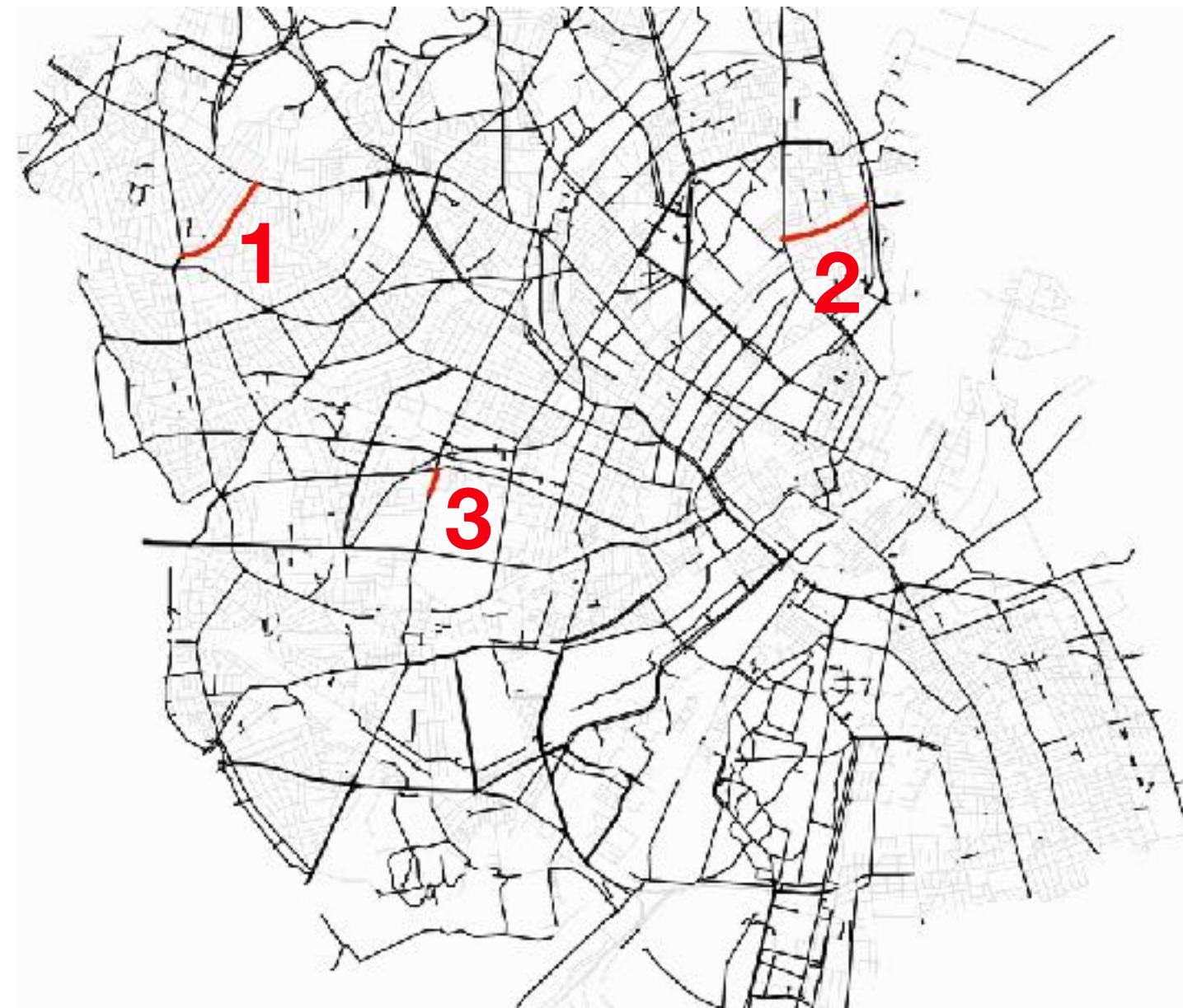


THE PROBLEM Cities struggle with bike network planning

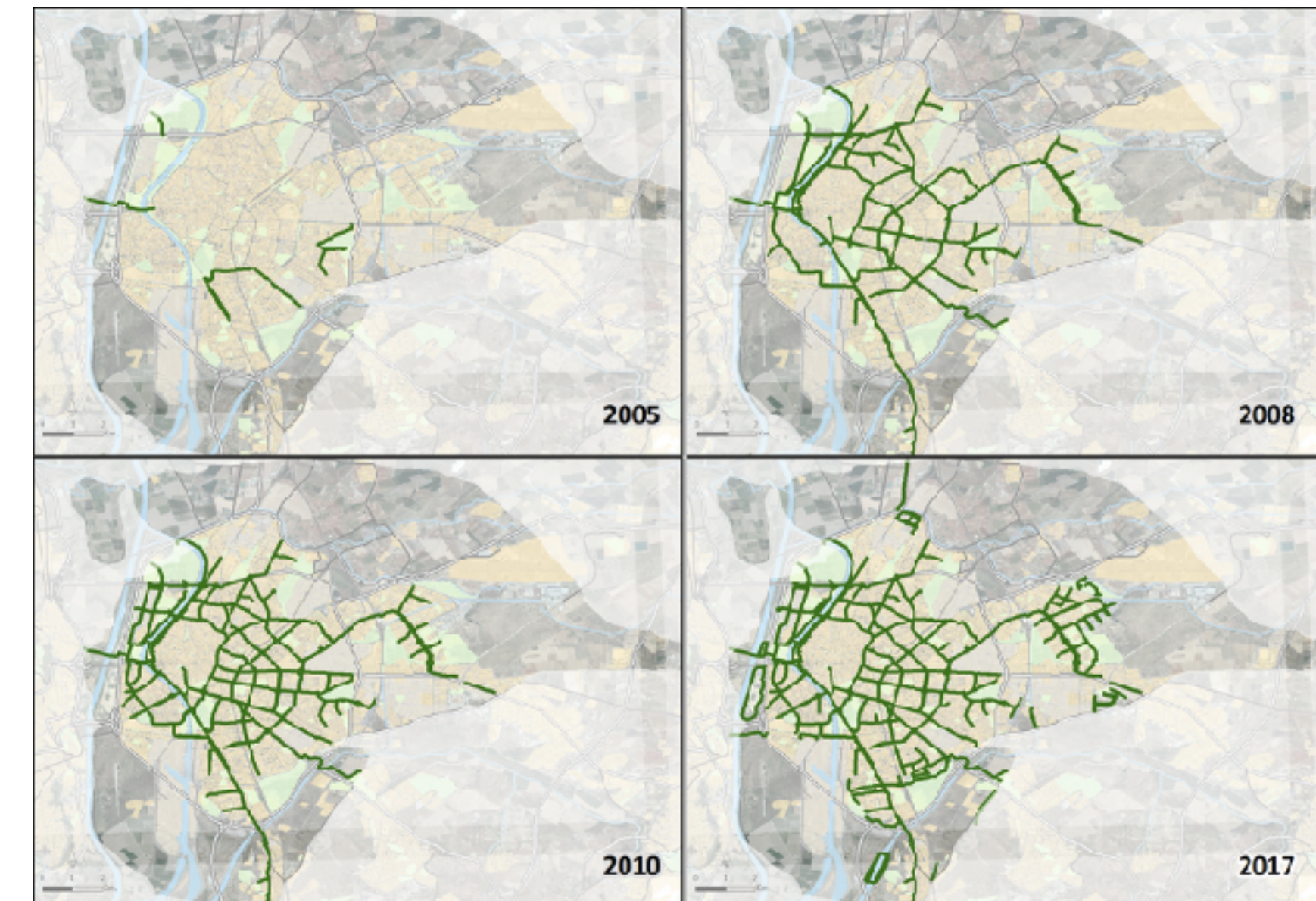
Current network planning is:

- manual
- political
- lacks success metrics

Where to
close the gaps?

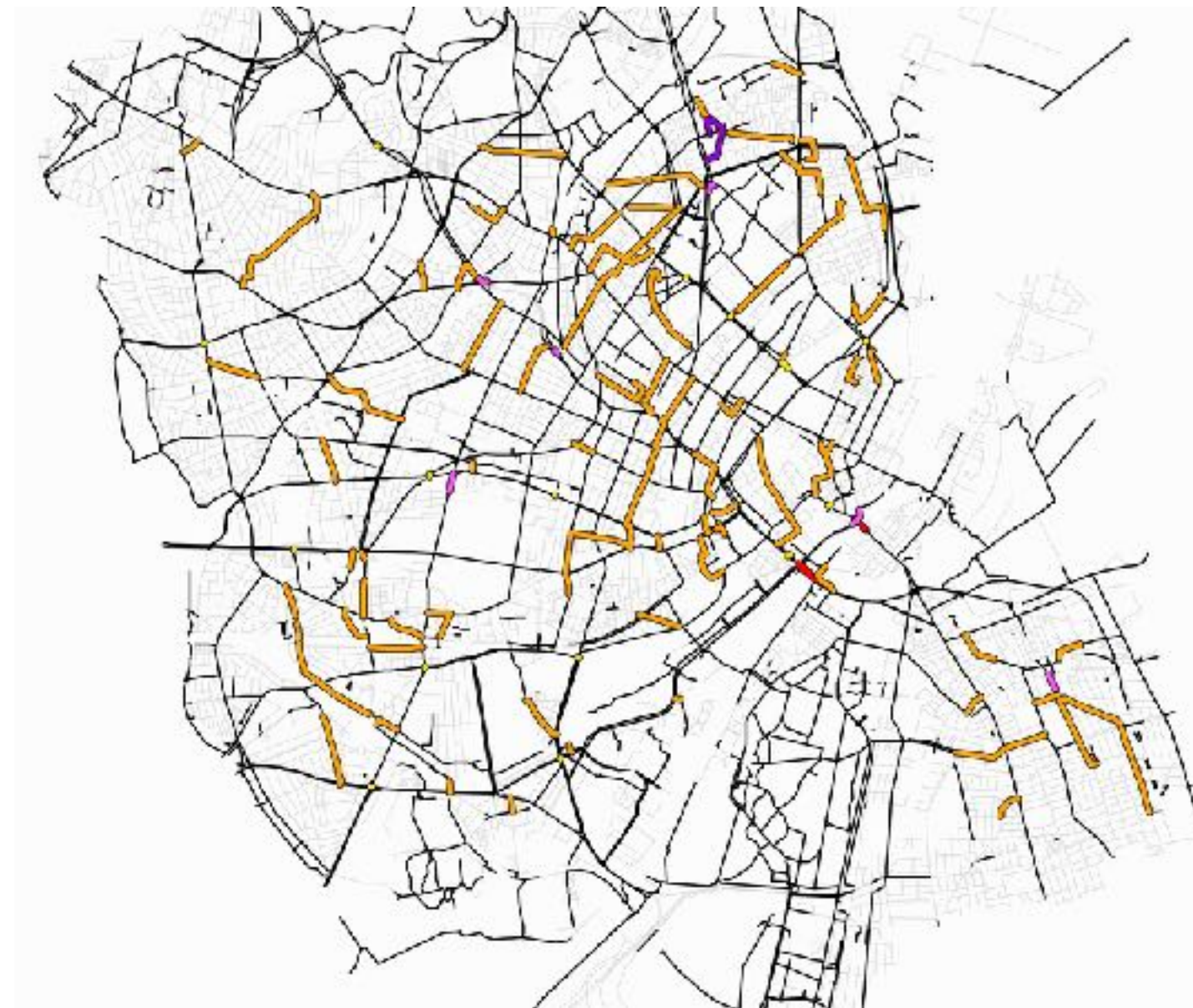


How to
build from scratch?

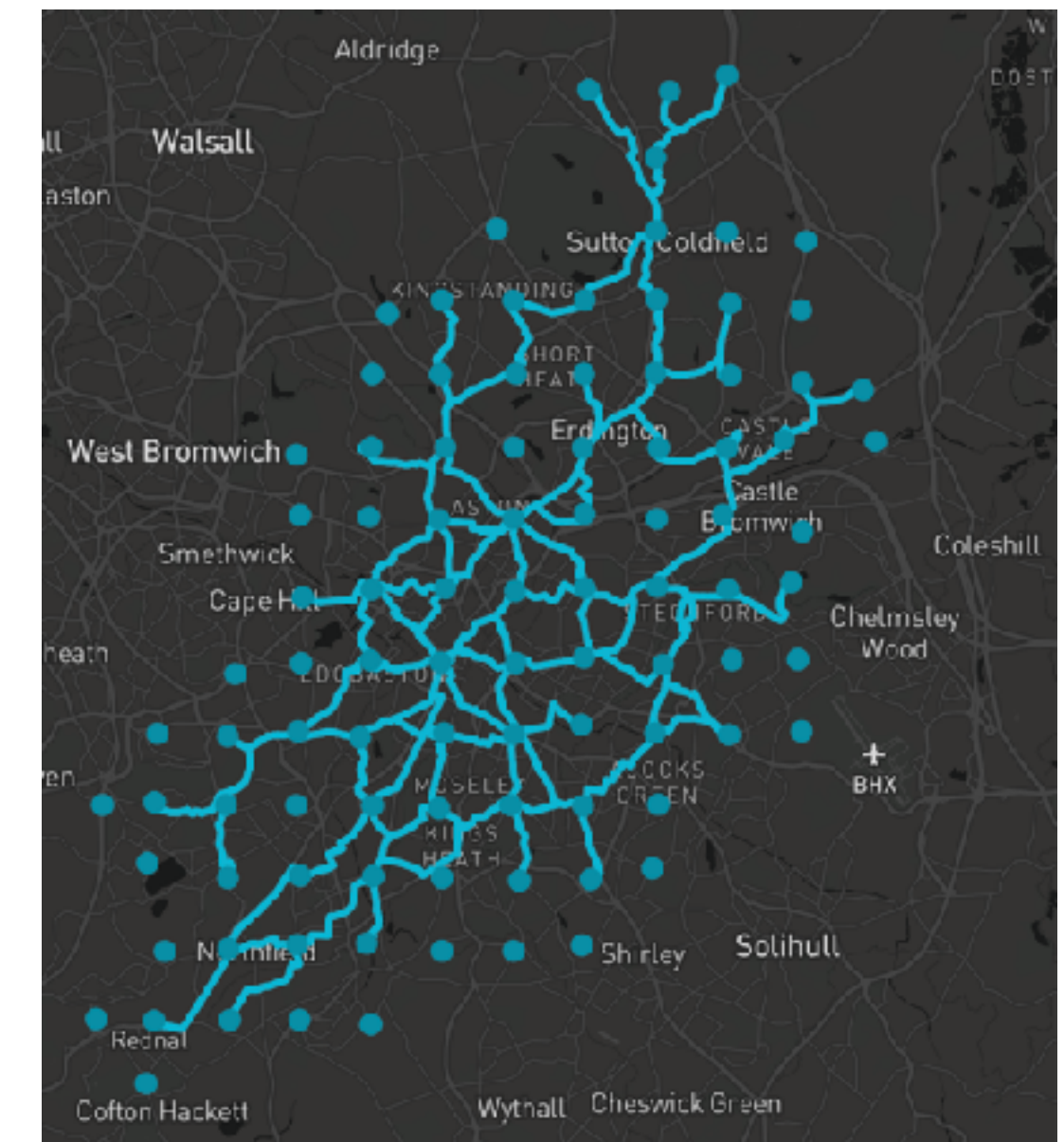


OUR SOLUTION Bicycle network algorithms, supporting planners

Here to
close the gaps!



This way to
build from scratch!



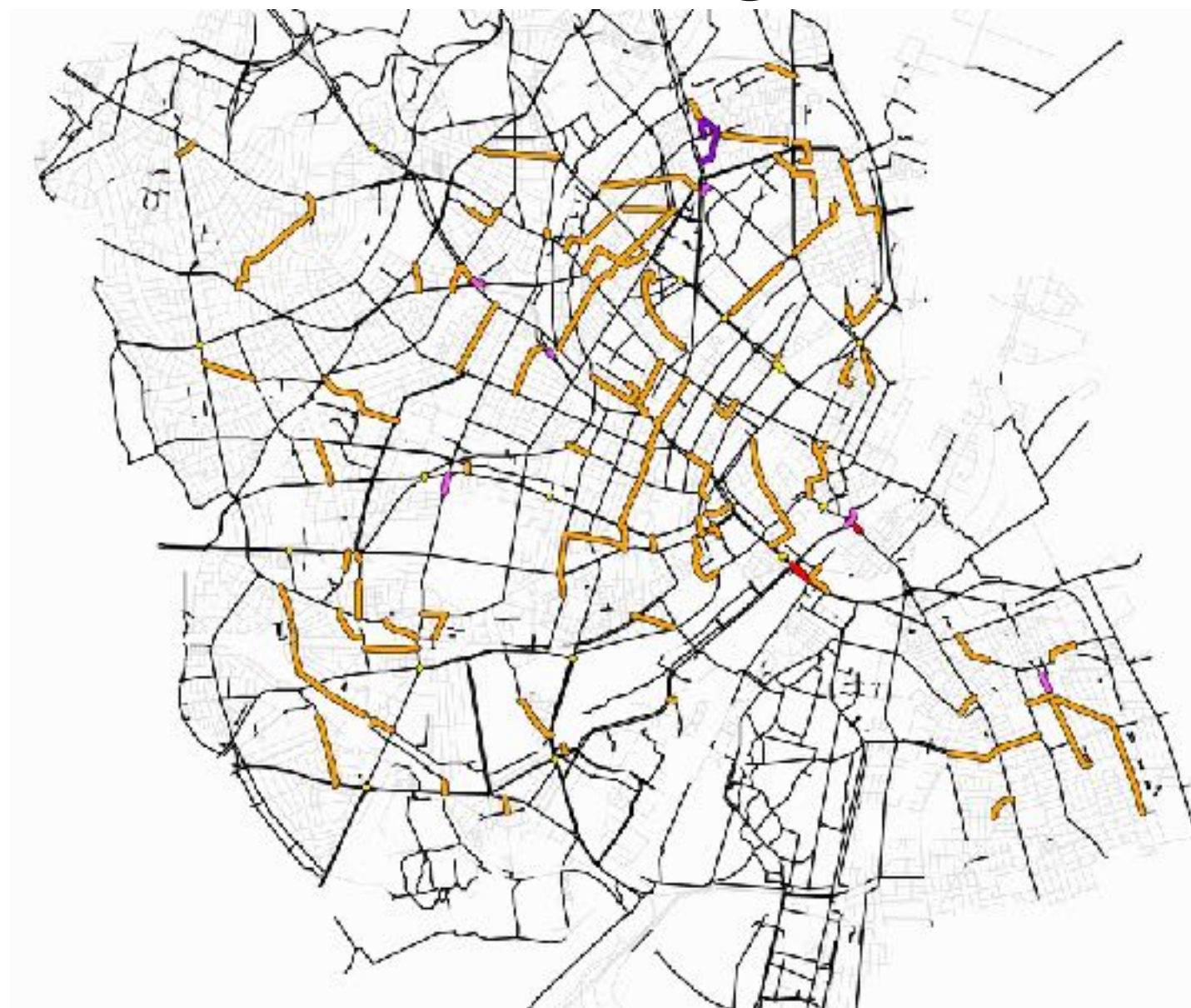
OUR SOLUTION Bicycle network algorithms, supporting planners

We have pioneered a
Science of Bicycle Networks

Key research/algorithms

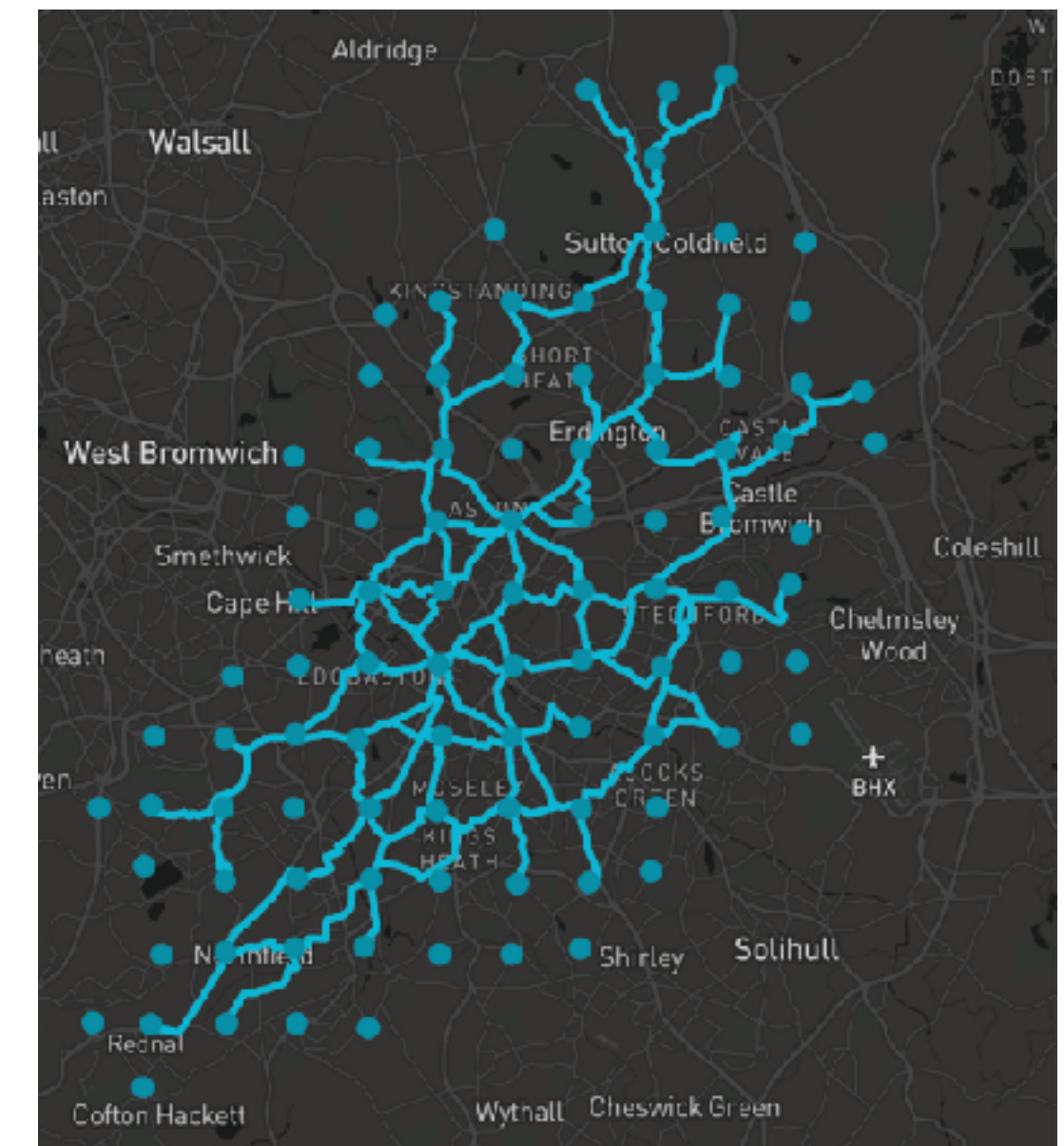
Sebastiao et al, Appl Net Sci (2026)
Szell et al, arxiv:2604.07029 (2026)
Vybornova et al, EPB 52 (2025)
Lonardi et al, R Soc Int 22 (2025)
Vierø et al, EPB 51 (2024)
Büth et al, JOSS 9 (2024)
Folco et al, EPB 50 (2023)
Vybornova et al, Geog Anal 55 (2023)
Szell et al, Sci Rep 12 (2022)
Breum et al, Transp F 56683 (2022)
Natera et al, R Soc Open Sci 7 (2020)

Here to
close the gaps!



FixBike.net

This way to
build from scratch!



GrowBike.net

We visualized 2 algorithms, exciting urban planners!

OUR PROBLEM Our algorithms are not yet user-friendly

- code
 - .ipynb_checkpoints
 - 01_prepare_networks.ipynb
 - 02_prepare_pois.ipynb
 - 03_poi_based_generation.ipynb
 - 04_analyze_results.ipynb
 - 05_plot_results.ipynb
 - 06_video_generation.ipynb
 - X_carconstricted_metrics.ipynb
 - X_export_data.ipynb
 - X_paper_figures_multicity.ipynb
 - X_sandbox.ipynb
 - X_supplements.ipynb
 - functions.py
 - path.py
 - setup.py
 - parameters

Download and wrangle data

Networks

```
In [ ]: for placeid, placeinfo in tqdm(cities.items(), desc = "Cities"):
        if placeinfo["nominatimstring"] != '':
            location = ox.geocoder.geocode_to_gdf(placeinfo["nominatimstring"])
            location = fill_holes(extract_relevant_polygon(placeid, shapely.geometry.shape(location['geometry'][0])))
            if debug: # Draw location polygons and their holes
                try:
                    color = cm.rainbow(np.linspace(0,1,len(location)))
                    for poly,c in zip(location, color):
                        plt.plot(*poly.exterior.xy, c = c)
                        for intr in poly.interiors:
                            plt.plot(*intr.xy, c = "red")
                except:
                    plt.plot(*location.exterior.xy)
                    plt.show()
            else:
                # https://gis.stackexchange.com/questions/113799/how-to-read-a-shapefile-in-python
                shp = fiona.open(PATH["data"] + placeid + "/" + placeid + ".shp")
                first = next(iter(shp))
                try:
                    location = Polygon(shapely.geometry.shape(first['geometry'])) # If shape file is given as linestring
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Gs = {}
for parameterid, parameterinfo in tqdm(osmnxparameters.items(), desc = "Networks", leave = False):
    for i in range(0,10): # retry
        try:
```


OUR PROBLEM Our algorithms are not yet user-friendly

```
code
└─ .ipynb_checkpoints
    01_prepare_networks.ipynb
    02_prepare_pois.ipynb
    03_poi_based_generation.ipynb
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Jupyter notebooks



Spaghetti code



OUR PROBLEM Our algorithms are not yet user-friendly

```
code
├── .ipynb_checkpoints
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├── 03_poi_based_generation.ipynb
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├── 06_video_generation.ipynb
├── X_carconstricted_metrics.ipynb
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├── X_paper_figures_multicity.ipynb
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├── X_supplements.ipynb
├── functions.py
├── path.py
├── setup.py
└── parameters
```

Download and wrangle data

Networks

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```

Jupyter notebooks

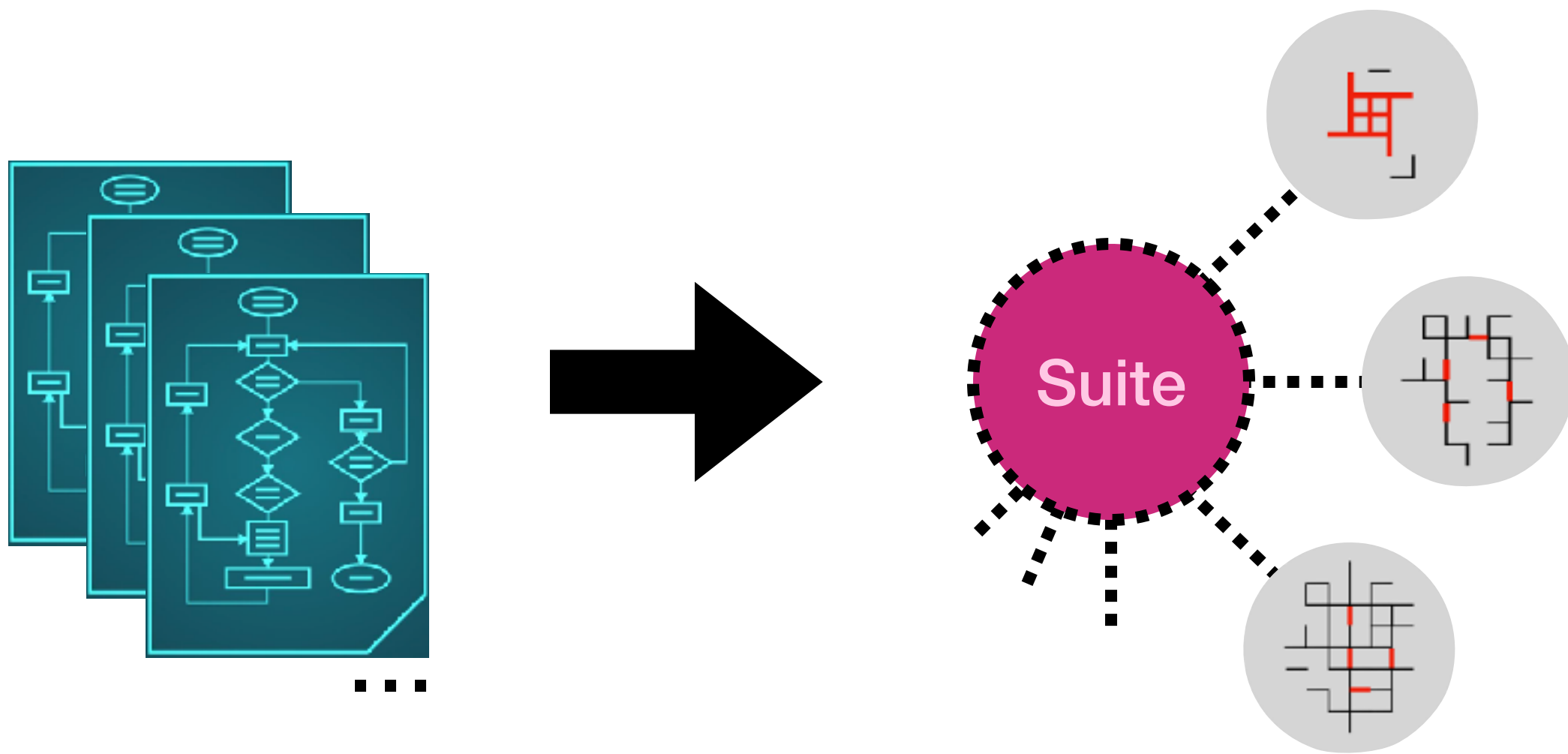


Spaghetti code



We got funds to 1) Create usable software

Innovation Fund Denmark

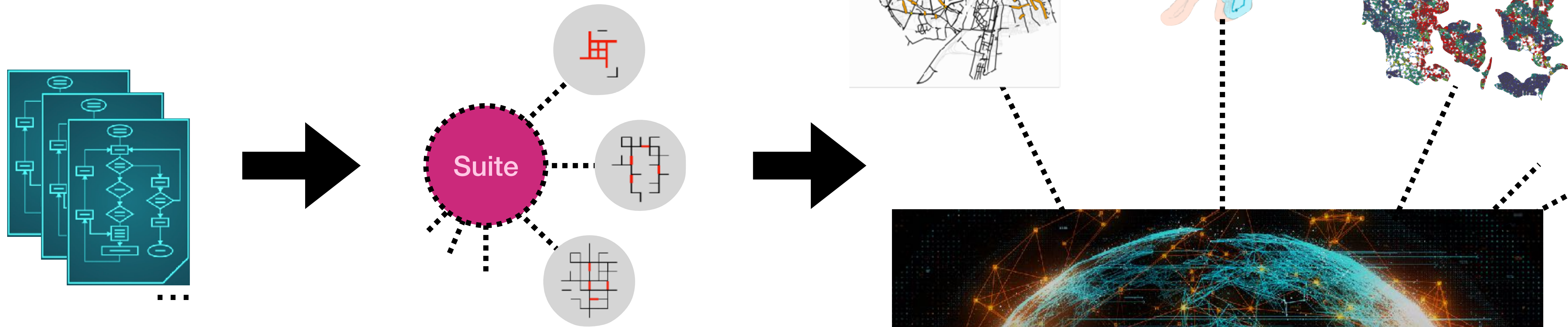


1) Research software engineer (Manuel)
creates packaged Python software

Usable by planners

We got funds to 1) Create usable software, 2) Visualize on the web

nnovation Fund Denmark



1) Research software engineer (Manuel) creates packaged Python software

Usable by planners

2) Web developer (Moritz) visualizes our solutions on a web platform

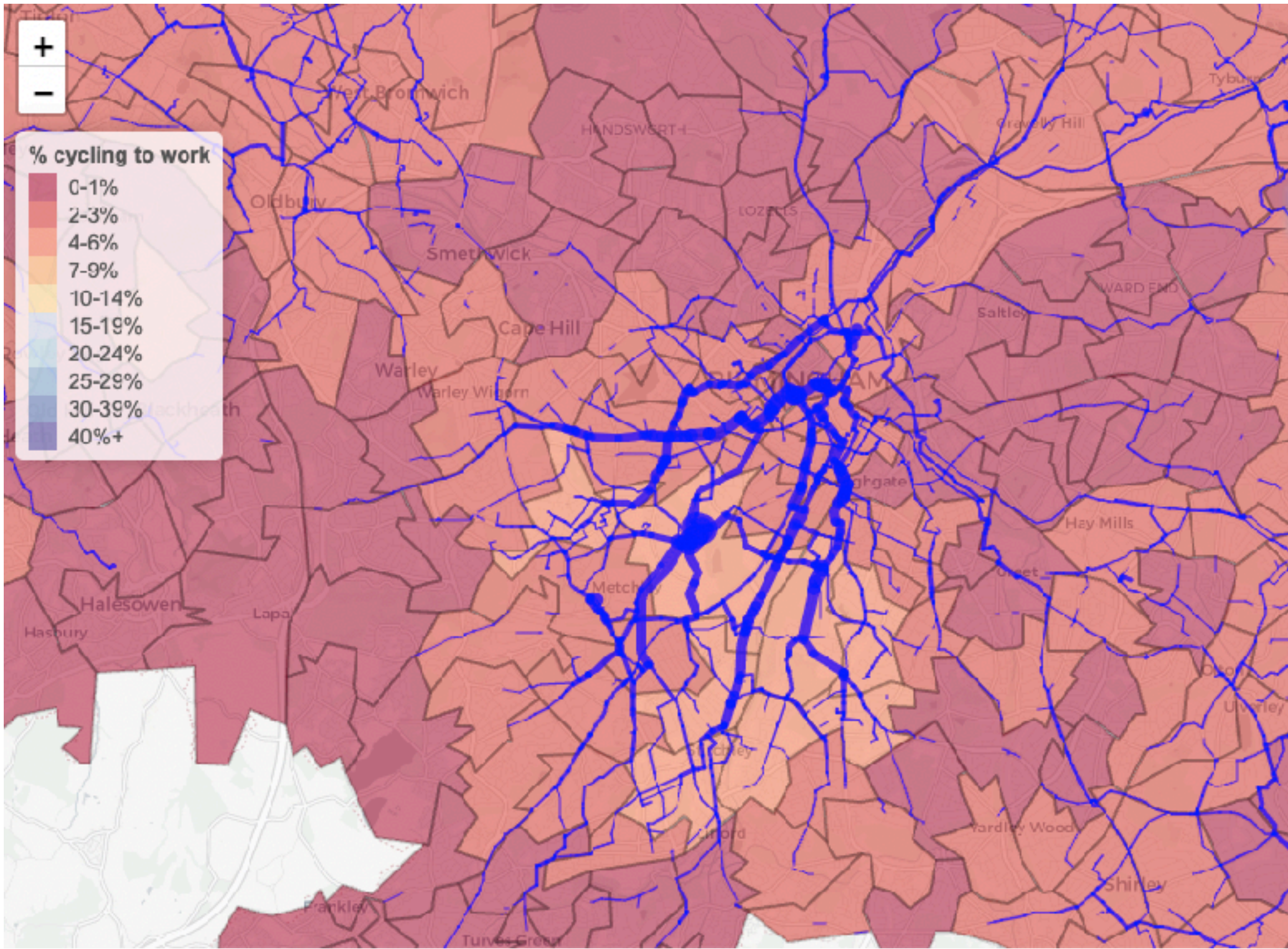
Guides planners, inspires citizens

We are not the first: Few fantastic free tools exist!

A/B Street

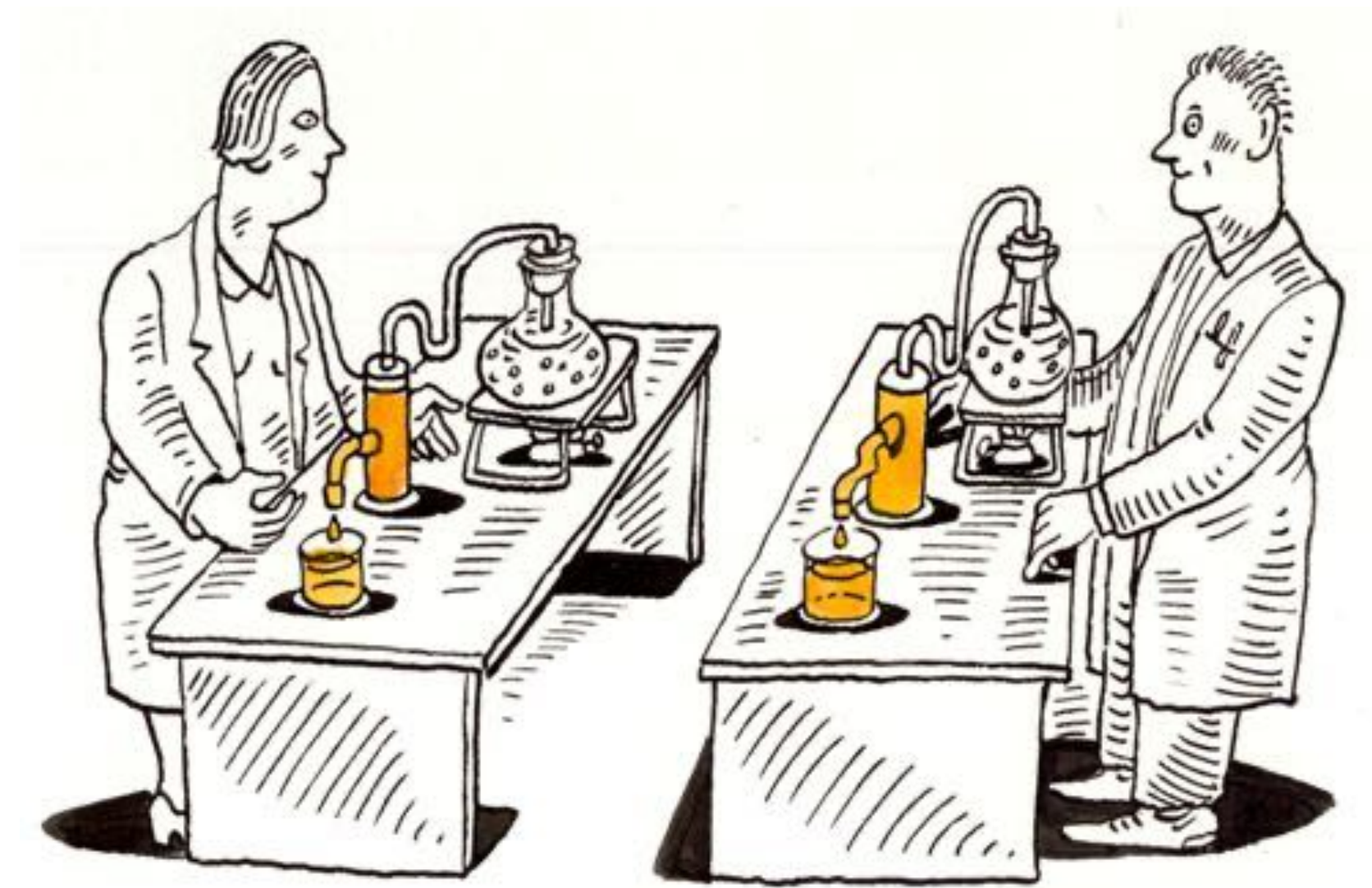


Propensity to Cycle Tool



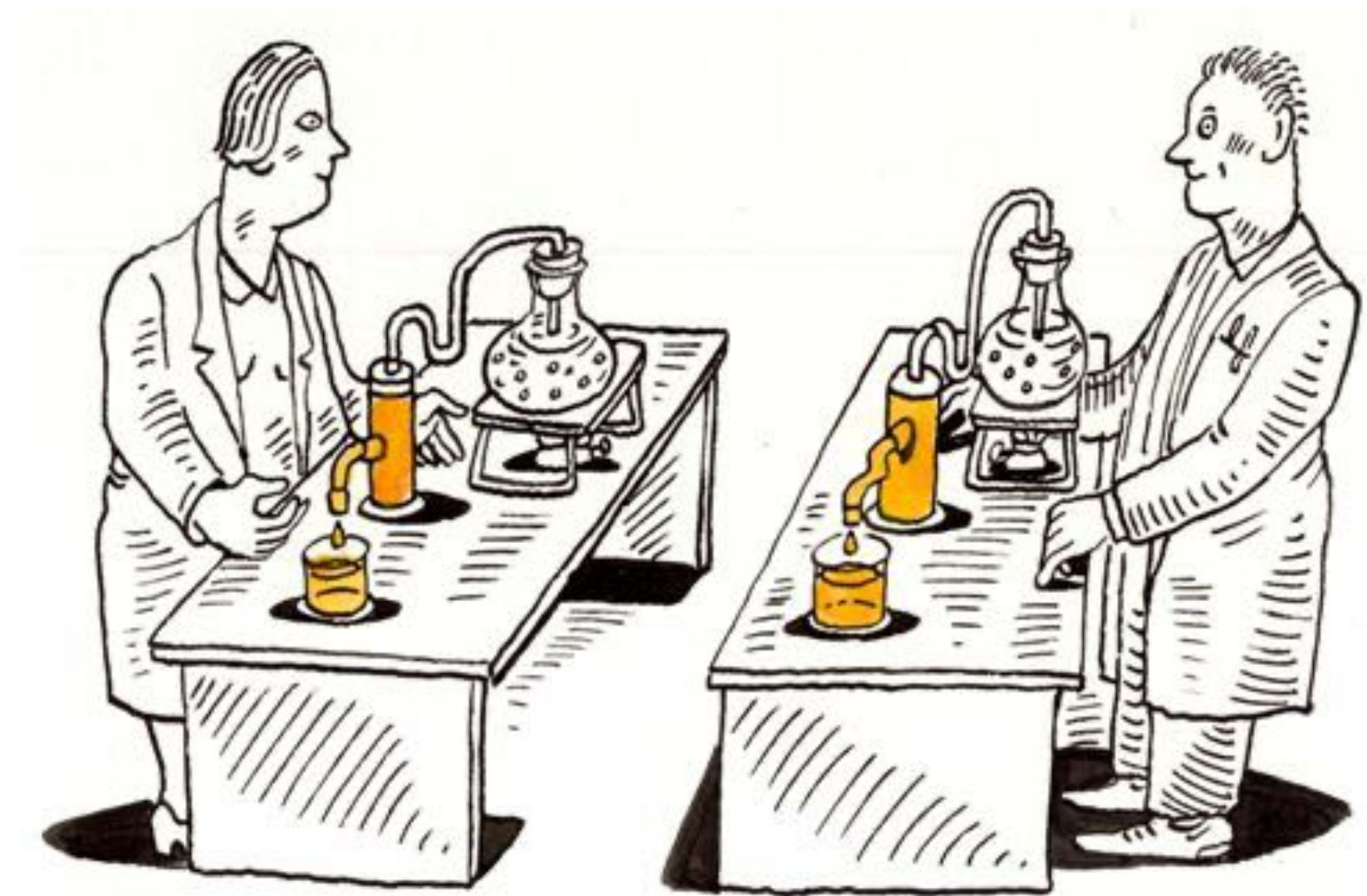
Our philosophy: Free, Open-Source, Reproducible

The software should be usable, modifiable, and reproducible by **anyone**, including proactive citizens, to create a compelling vision for urban cycling in their city.



Our philosophy: Free, Open-Source, Reproducible

The software should be usable, modifiable, and reproducible by **anyone**, including proactive citizens, to create a compelling vision for urban cycling in their city.



It should be an **alternative to extractive practices** where cities overpay consultants for a one-off report



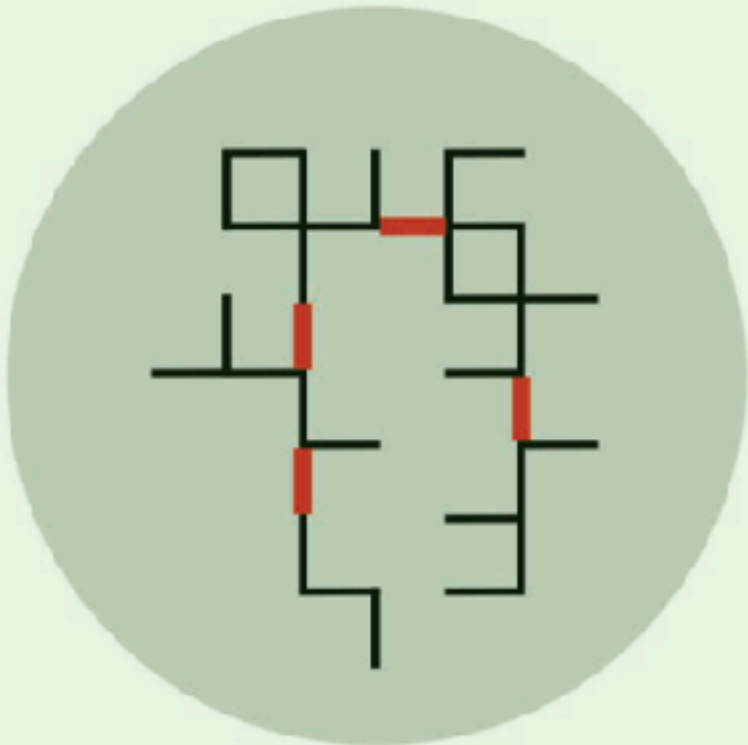
Our tools

We are creating a Python software suite: bikenetkit.org

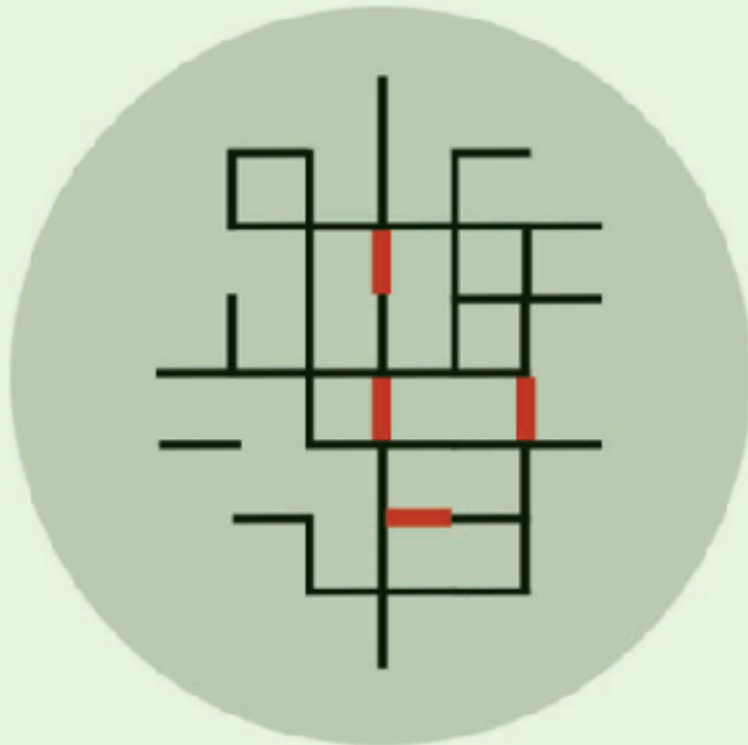
GrowBikeNet



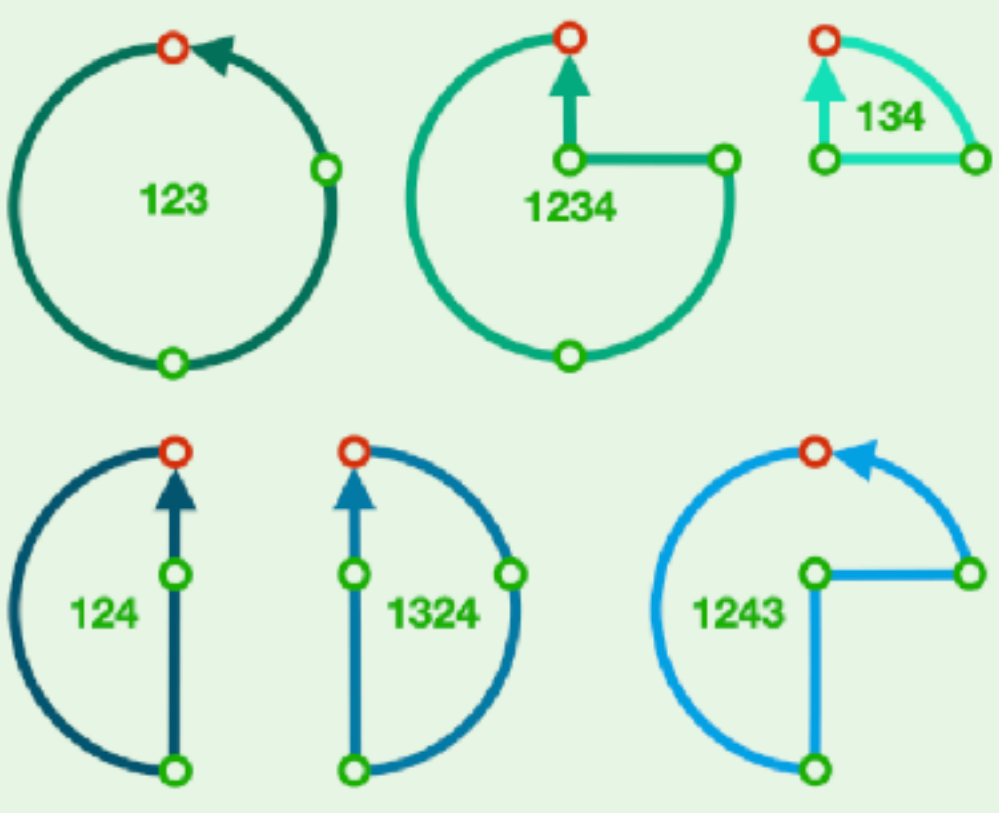
LinkBikeNet



FixBikeNet



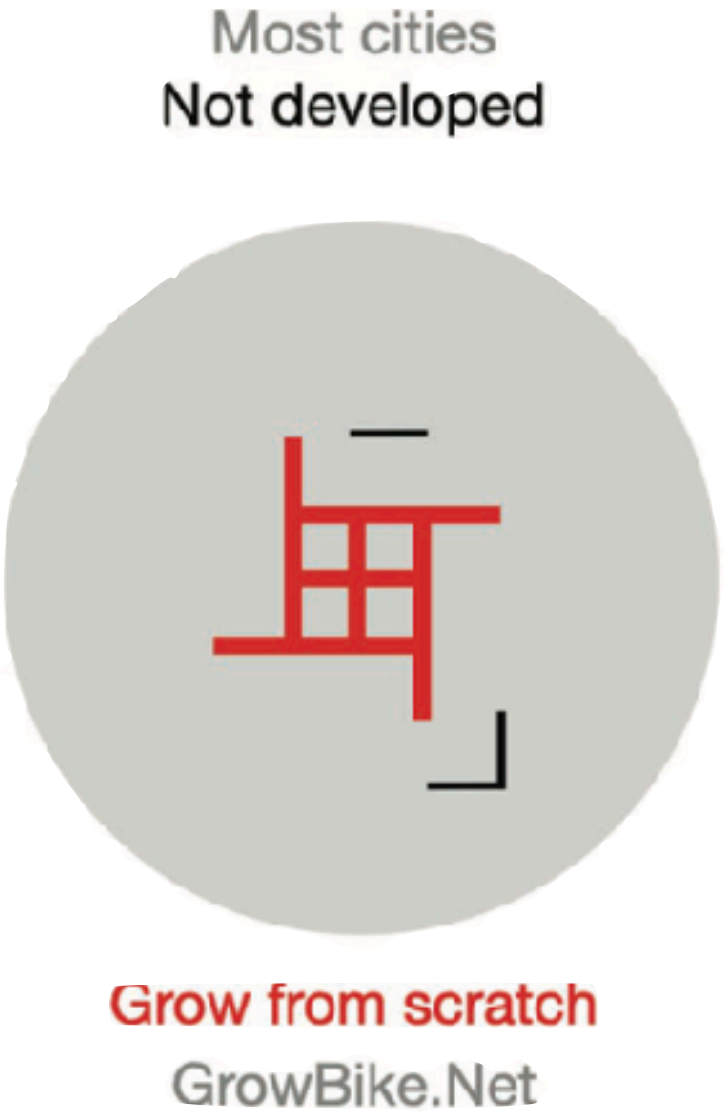
LoopBikeNet



ENPERHLLCHIFY

Different cities need different network design approaches

GrowBikeNet



Different cities need different network design approaches

GrowBikeNet

Birmingham

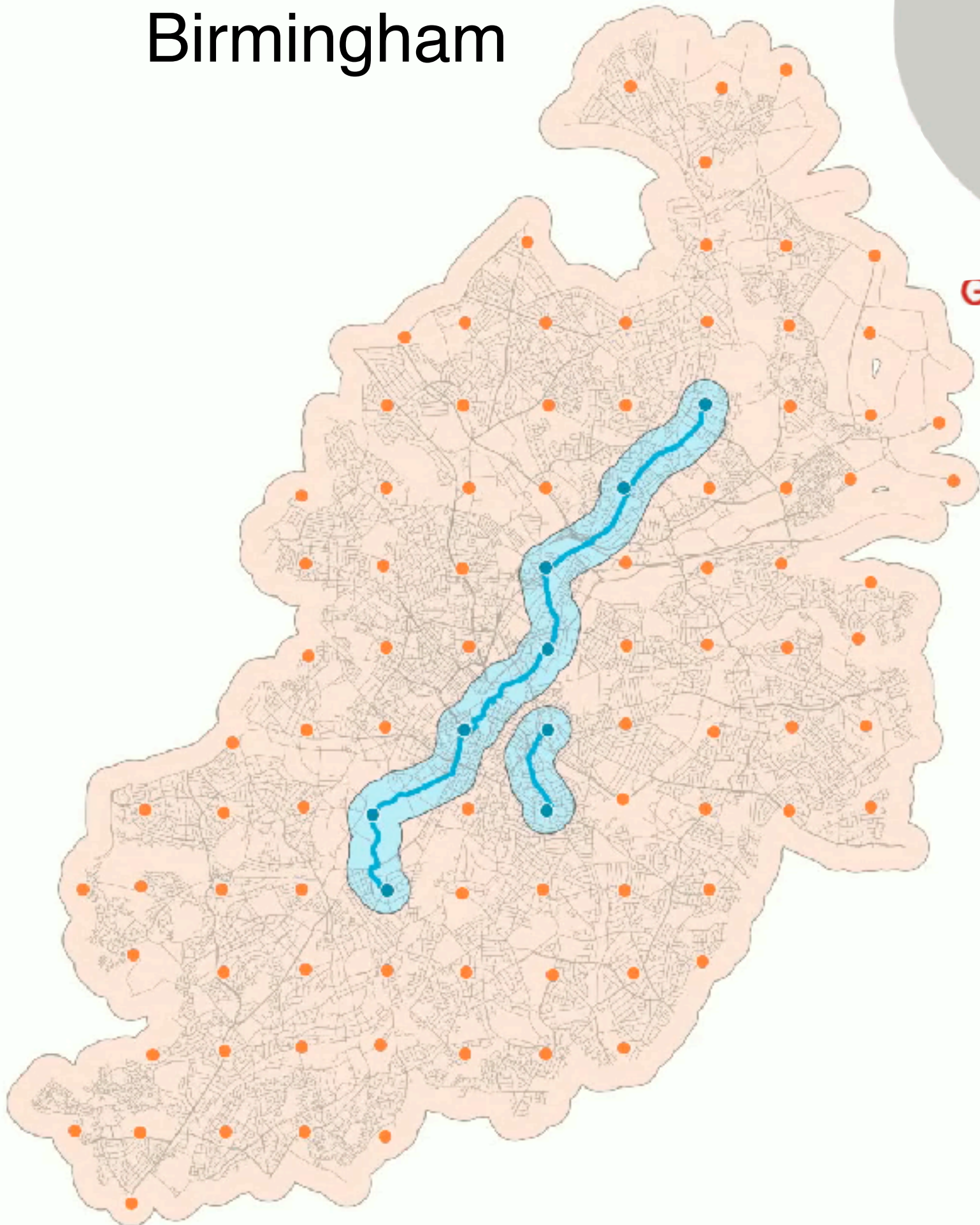
Most cities
Not developed



Grow from scratch
GrowBike.Net

Different cities need different network design approaches

Birmingham

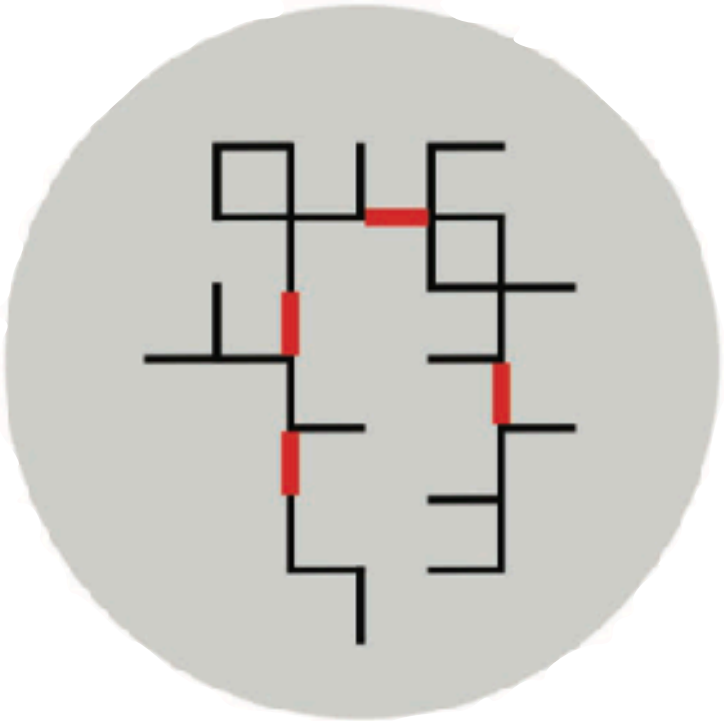


Most cities
Not developed



Grow from scratch
GrowBike.Net

Some cities
Developed but
disconnected



Connect components
LinkBike.Net

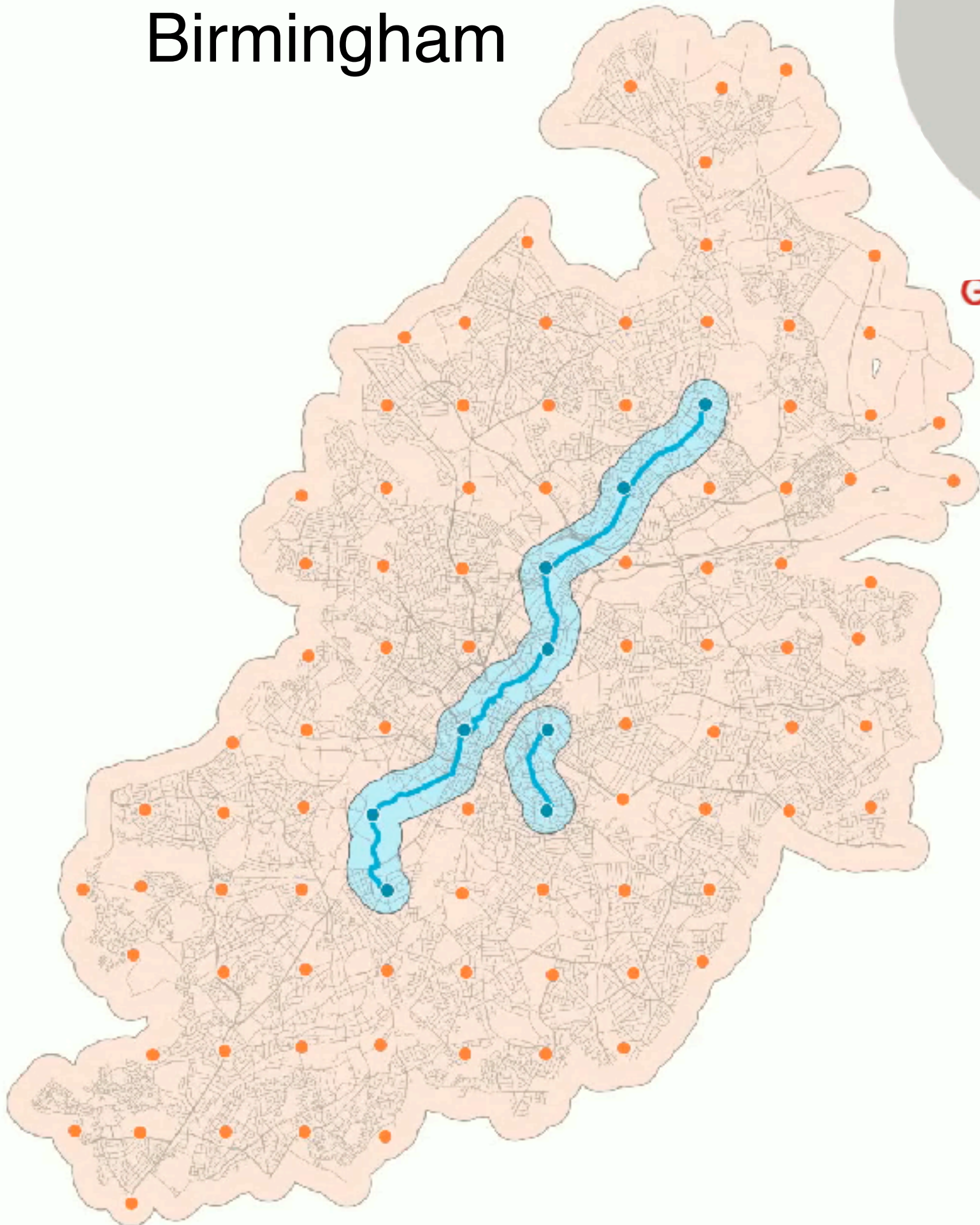
Budapest



Different cities need different network design approaches

FixBikeNet

Birmingham

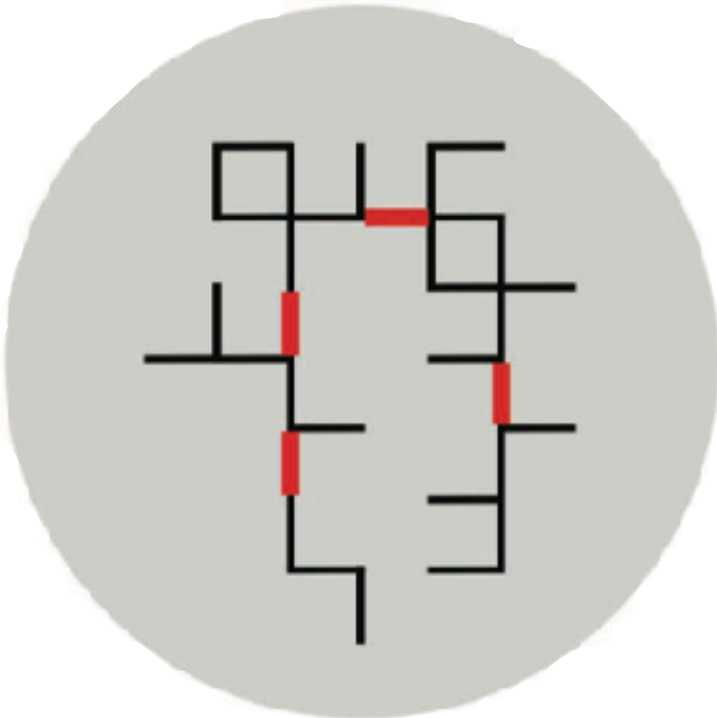


Most cities
Not developed



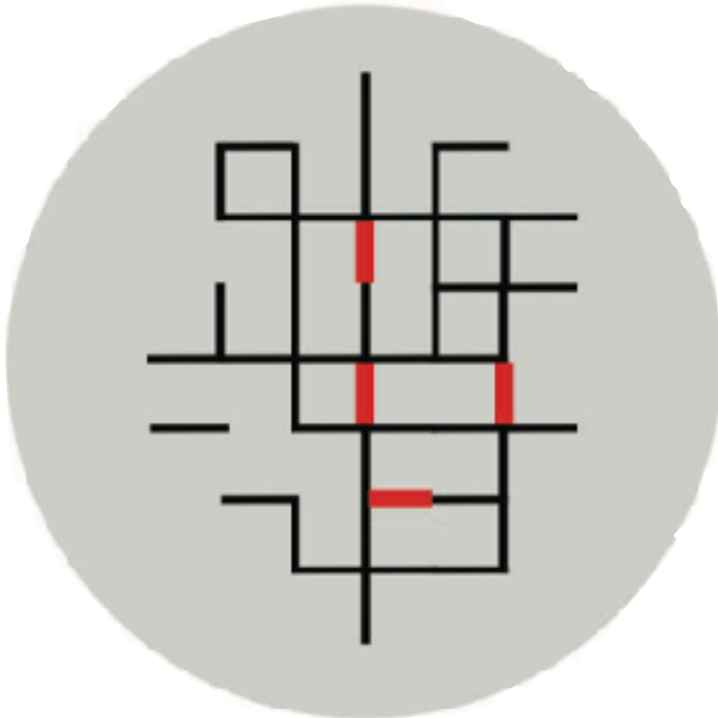
Grow from scratch
GrowBike.Net

Some cities
Developed but
disconnected



Connect components
LinkBike.Net

Few cities
Developed and
mostly connected



Find missing links
FixBike.Net

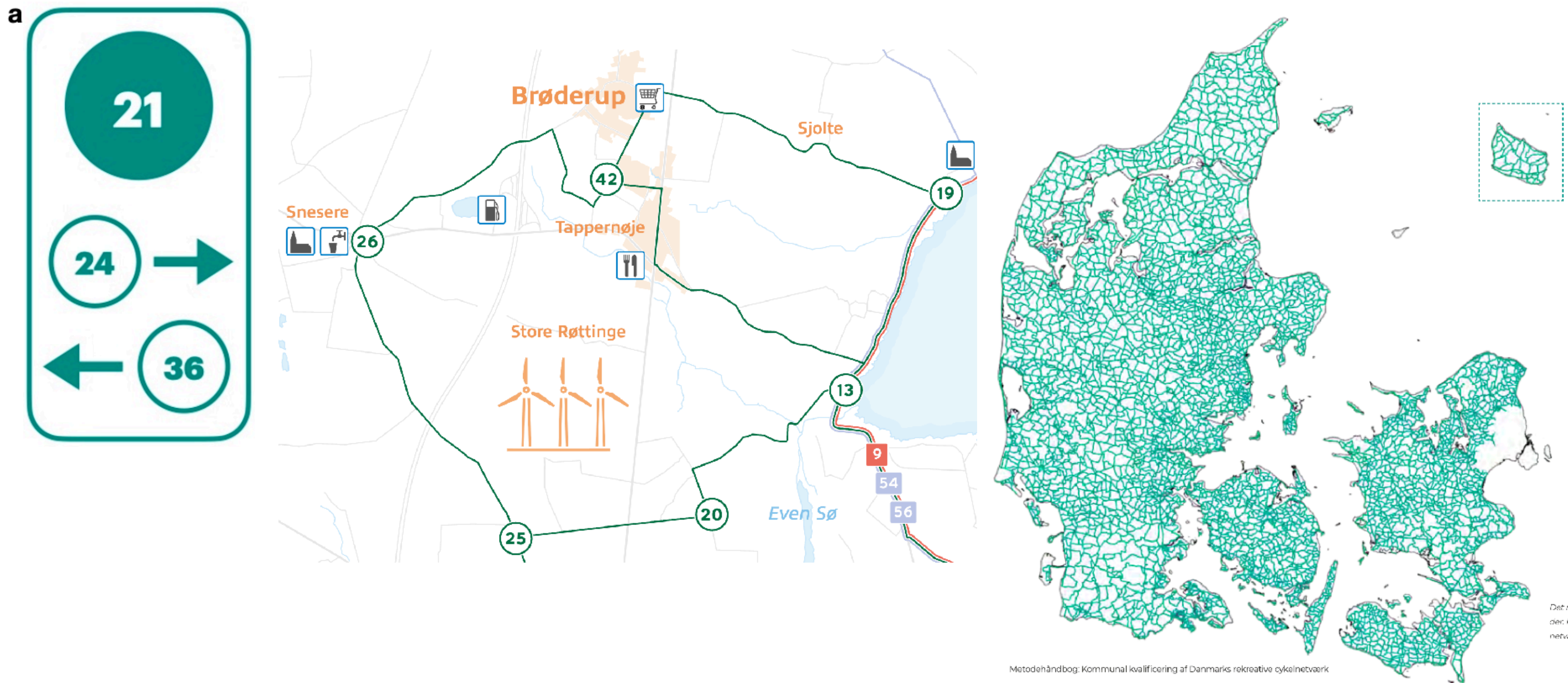
Budapest



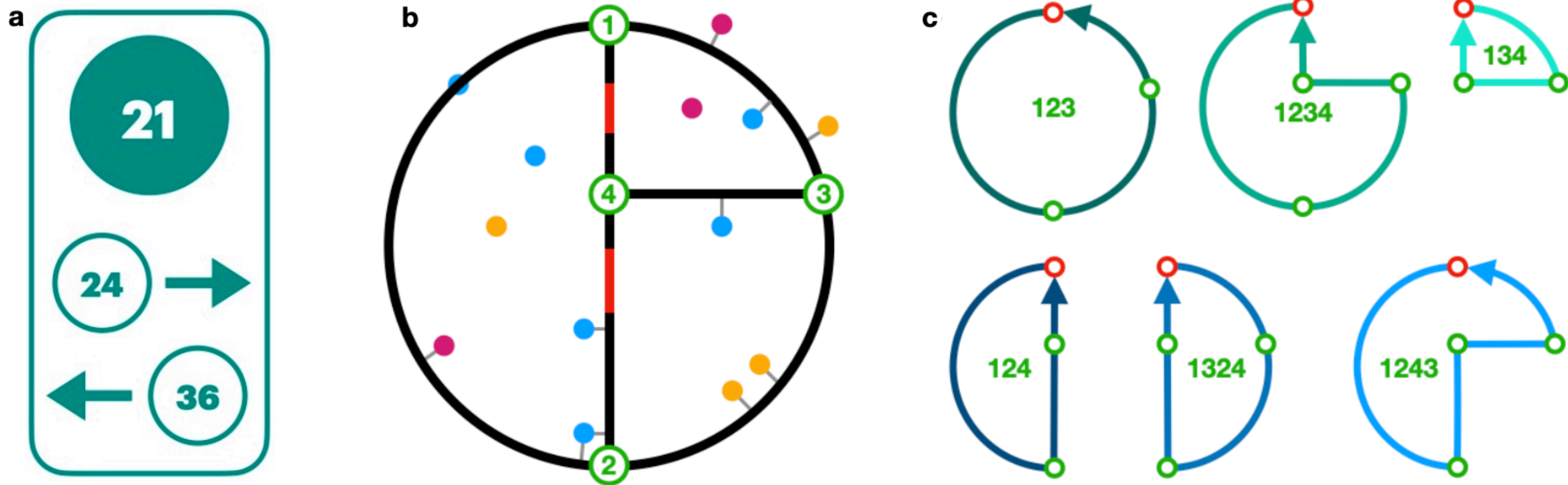
Copenhagen



Regional, recreational networks are a different beast



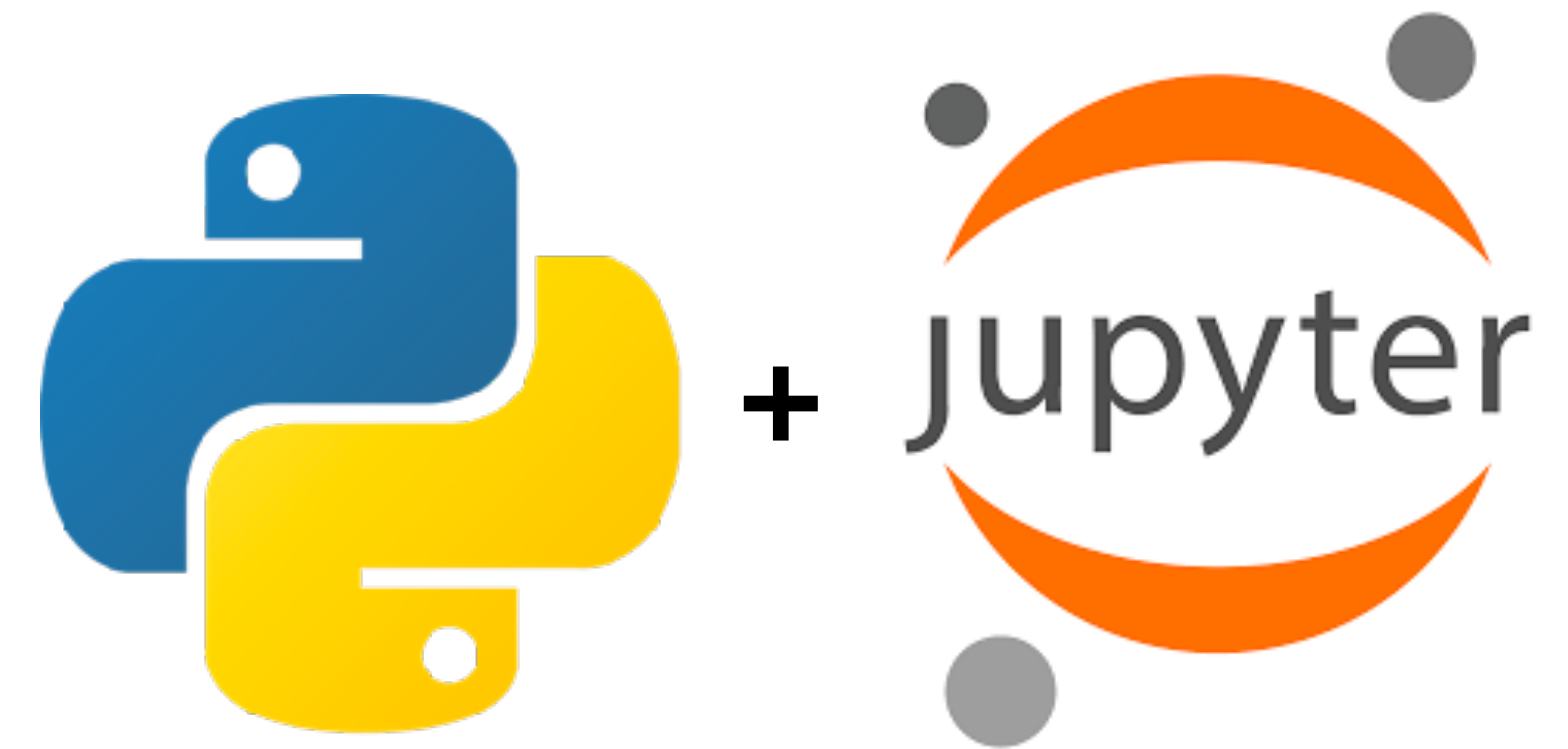
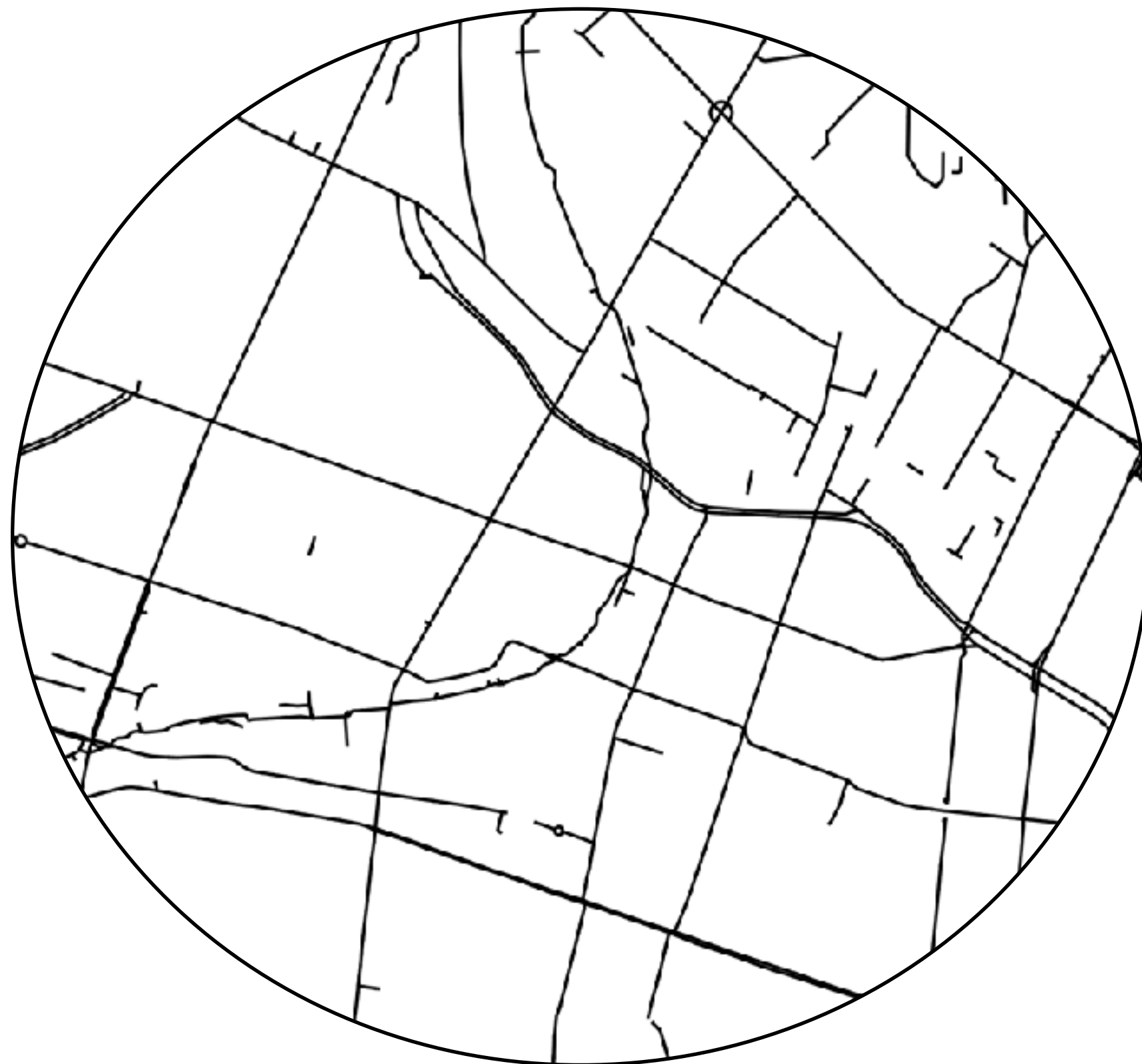
Regional, recreational networks are a different beast



Loop census: All roundtrips from one node

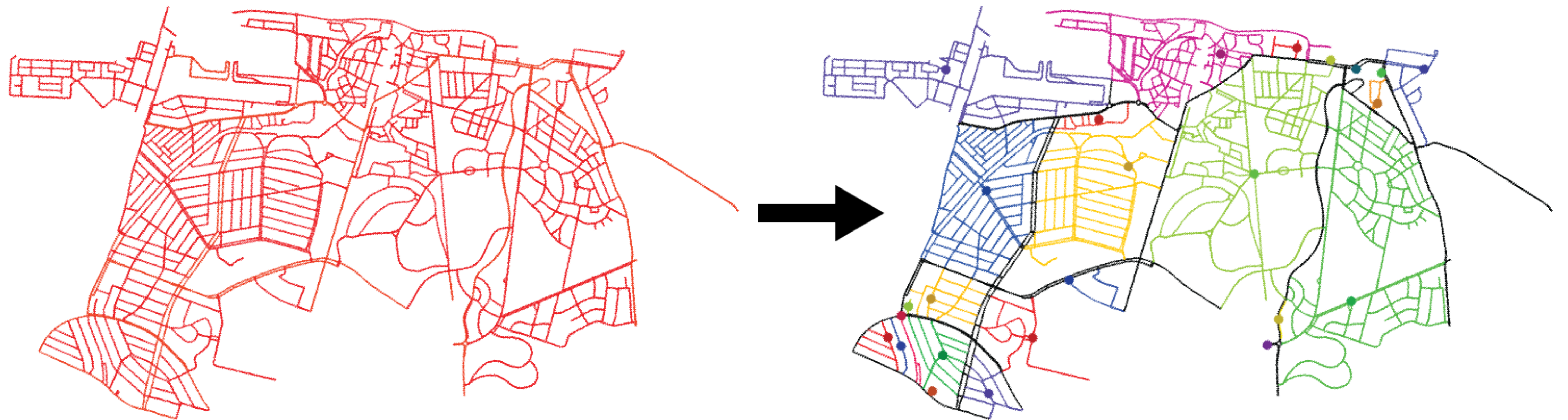


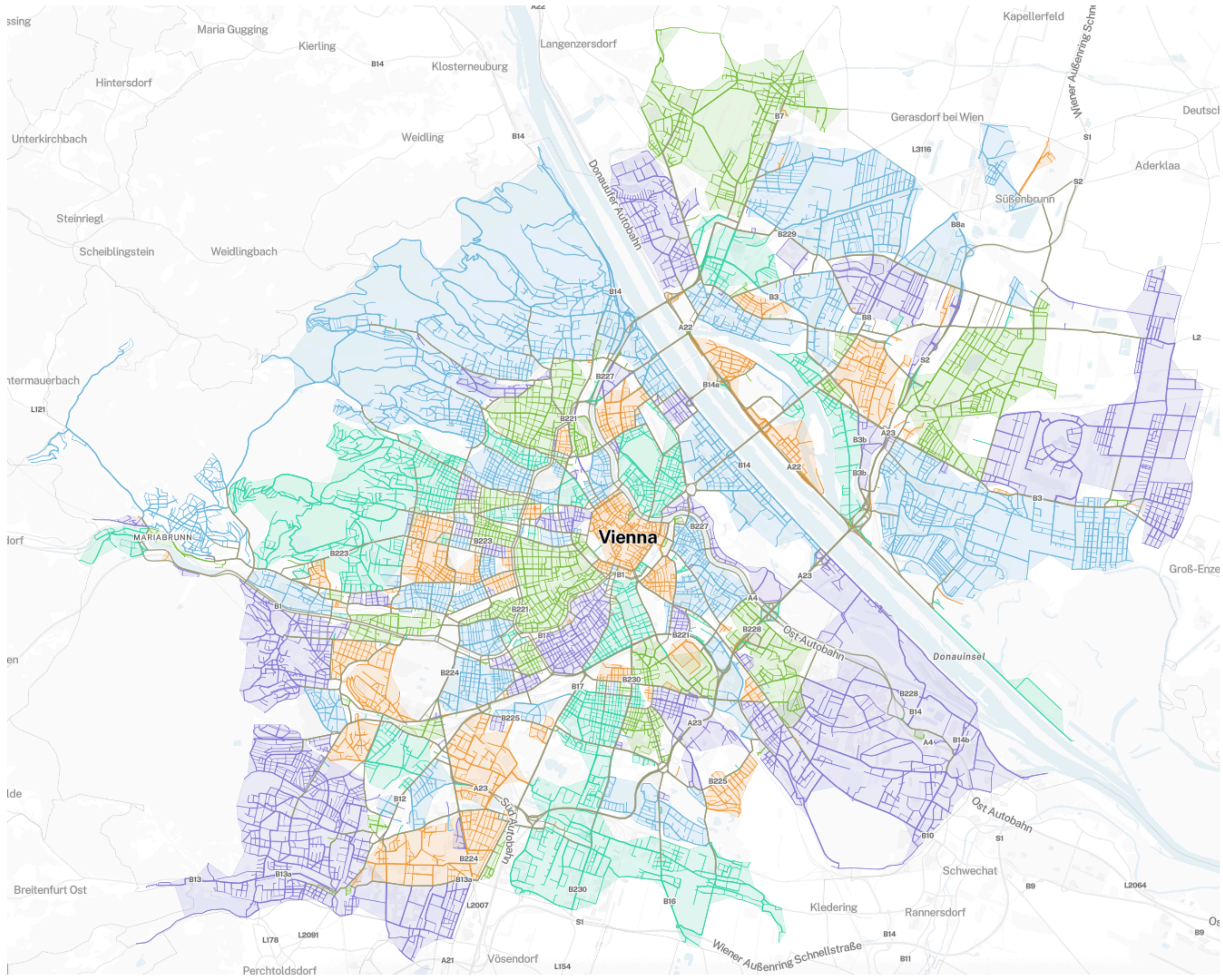
Bicycle Infrastructure Data & Network Assessment



Funded by The Danish Road Directorate 

SUPERBLOCKIFY





Our progress

The core team (so far)

Software development

	<u>Carlson M. Büth</u>		<u>Manuel Knepper</u>		<u>Mariana Pessoa</u>
	<u>Michael Szell</u>		<u>Anastassia Vybornova</u>		

Design & Visualization

	<u>Moritz Stefaner</u>
---	--

Lead

	<u>Michael Szell</u>
---	--------------------------------------

Development status

Tool	Status	Version
superblockify	 Stable	1.0.2
GrowBikeNet	 Beta	0.10.3
FixBikeNet	 Alpha	0.7.0
LinkBikeNet	 Planning	n/a
LoopBikeNet	 Projected	n/a
BikeNetLib	 Planning	0.5.0

Live Demo at

<https://bikenetkit.netlify.app/>

Live Demo at

<https://bikenetkit.netlify.app/>

Until released (~Dec 2026)
we show a preliminary screen at

bikenetkit.org



BikeNetKit

The digital toolkit for bicycle networks

Free, open-source Python tools for designing well-connected bicycle infrastructure.

```
[BIKENETKIT_V0.4.0]
> CITY : Rome, Italy
> ALGO : growbikenet
> METRIC: closeness centrality
> SEED : rail
```

```
> growing network... 90%
FRAME 0361 / 0402
LENGTH 925.6 / 1145.3 km
```

```
LAT 41.8761° N
LON 12.5279° E
BRG 247°
PIT 050°
ZOOM 9.15
```

// bikenetkit |



Coming soon!
Meanwhile, follow
us on →



Bluesky



GitHub



Mastodon

/nnovation Fund Denmark

SUPPORTED BY



Challenges: Project management

Hiring a good web dev / designer within <4 months

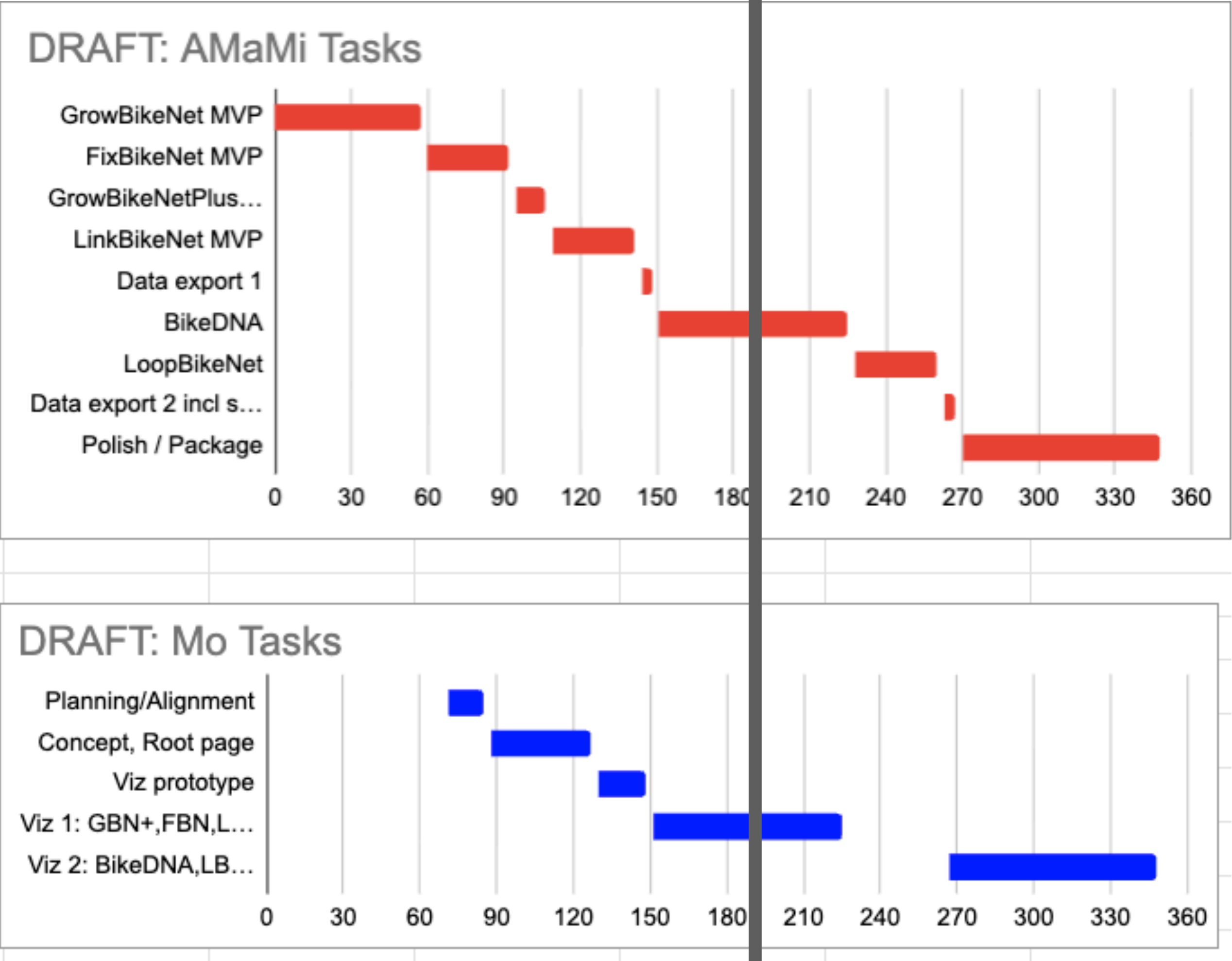


Challenges: Project management

Set up a concrete plan, coordinate team

We are here

Who	Start	End	Task
AMaMi	2026-01-01	2026-02-27	GrowBikeNet MVP
AMaMi	2026-03-02	2026-04-03	FixBikeNet MVP
MaMi	2026-04-06	2026-04-17	GrowBikeNetPlus MVP
MaMi	2026-04-20	2026-05-22	LinkBikeNet MVP
MaMi	2026-05-25	2026-05-29	Data export 1
MaMi	2026-06-01	2026-08-14	BikeDNA
MaMi	2026-08-17	2026-09-18	LoopBikeNet
MaMi	2026-09-21	2026-09-25	Data export 2 incl superblocify
MaMi	2026-09-28	2026-12-15	Polish / Package
			Task
Mo	2026-03-13	2026-03-27	Planning/Alignment
Mo	2026-03-30	2026-05-08	Concept, Root page
Mo	2026-05-11	2026-05-29	Viz prototype
Mo	2026-06-01	2026-08-14	Viz 1: GBN+,FBN,LBN
Mo	2026-09-25	2026-12-15	Viz 2: BikeDNA,LBN,superblockify



Backlog

16 / 5

Estimate: 0

...

+

This item hasn't been started

GrowBikeNet #176

Find an alternative to openCV for movie creation

P1

GrowBikeNet #61

Implement alpha between two data sets

P2

GrowBikeNet #83

After triangulation bugfix, add minimum weight triang back as option

P2

GrowBikeNet #94

Once data import is implemented, add more test cases

P2

GrowBikeNet #140

Improve export folder structure

P2

GrowBikeNet #163

Improve speed of OSMnx retrieval from cache

Ready

4

Estimate: 0

...

+

This is ready to be picked up

GrowBikeNet #187

Move all constants/settings to constants/settings

P1

GrowBikeNet #60

Implement trip data to edge matching

P2

GrowBikeNet #149

Update and test new example export scripts from dataexport

GrowBikeNet #154

Clean up minor docstring and duplicate citation issues

In progress

2 / 3

Estimate: 0

...

+

This is actively being worked on

GrowBikeNet #59

Implement node data to edge matching

P2

GrowBikeNet #155

Add more usage examples to docs

#166 #169

In review

0 / 5

Estimate: 0

...

+

This item is in review

Done

71

Estimate: 0

...

+

This has been completed

GrowBikeNet #185

Make loading bike network possible

#189 P0

GrowBikeNet #147

Update custom filter

#148 P0

GrowBikeNet #114

Edge overlaps are not completely removed

#120 P0

GrowBikeNet #106

Batch data export for all european capitals

P0

GrowBikeNet #102

Add new parameter to allow graph from polygon instead of nominatim query

#108 P0

GrowBikeNet #103

After graph from polygon implemented, package 0.7.0

P0

Challenges: Software development

Learning+handling all the technicalities of software development

- **Github**: branches, PRs, issues, projects, actions, pre-commit hooks, badges
- **Packaging**: PyPI, conda-forge, releases
- **Testing**: pytest
- **Docs**: Sphinx, caching

Not using any AI, as we needed to actually learn all this!

Many reasons against AI

Legal/ethical: Copyright issues

Sustainability: Energy/water consumption

Quality: Lot of garbage

Energy vampire: Removes valuable reviewer time

PRs are good for mentoring and training future core contributors. But AI PRs are always "drive-by" contributions.



<https://www.youtube.com/watch?v=iqddnwKF8HQ>

How can you help?

- Like/Star it on Github: <https://github.com/BikeNetKit>
- Follow us / post about it on social media. We have social media accounts on 3 platforms: 
- Try out the software. Let us know what works/doesn't.
- Help with "good first issue" in Github issues
- Visit us in Copenhagen @ NERDS



Networks, Data, and Society

Research group at IT University of Copenhagen



We are researching network and data science applications to social systems

We are interdisciplinary: Data Science, Physics, Mathematics, GIS,...

We are alumni of MIT, Harvard, Cornell, Northeastern, Bell Labs, UN

We won the Young Academy's prize Research Environment of the Year 2022

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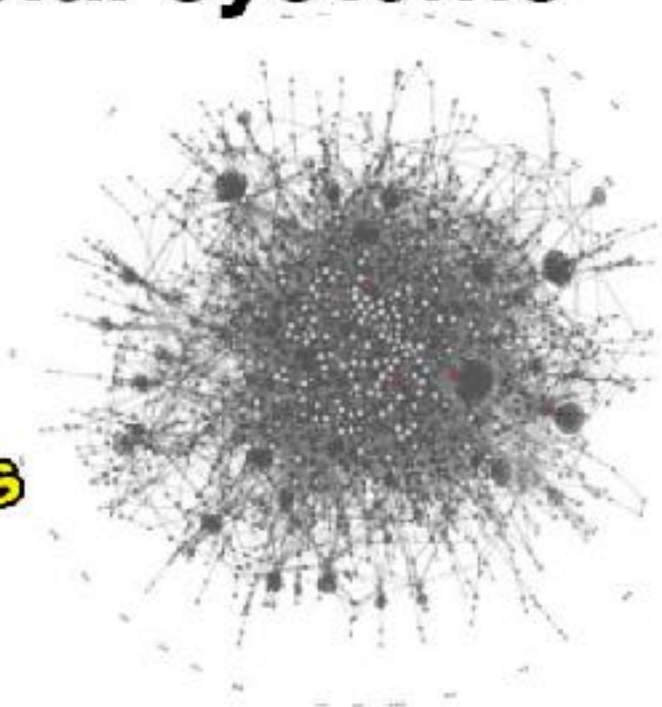
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Social Networks
Bicycle Networks
Complex Systems
NLP



All our data and code are open and reproducible



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Give us critical feedback, help us improve the software!

<https://github.com/BikeNetKit>

